

Stop-and-Go Traffic Wave Suppression Strategies

Variable Speed Limit vs. Jam-Absorption Driving

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A Review of Stop-and-Go Traffic Wave Suppression Strategies: Variable Speed Limit vs. Jam-Absorption Driving

Zhengbing He, Jorge Laval, Yu Han, Andreas Hegyi, Ryosuke Nishi, Cathy Wu



Stop-and-Go Waves on Freeways

Main Form: Stop-and-go waves (also wide moving jams, shock waves, or traffic oscillations)

(a) Nashville, TN, USA: 6.7 km

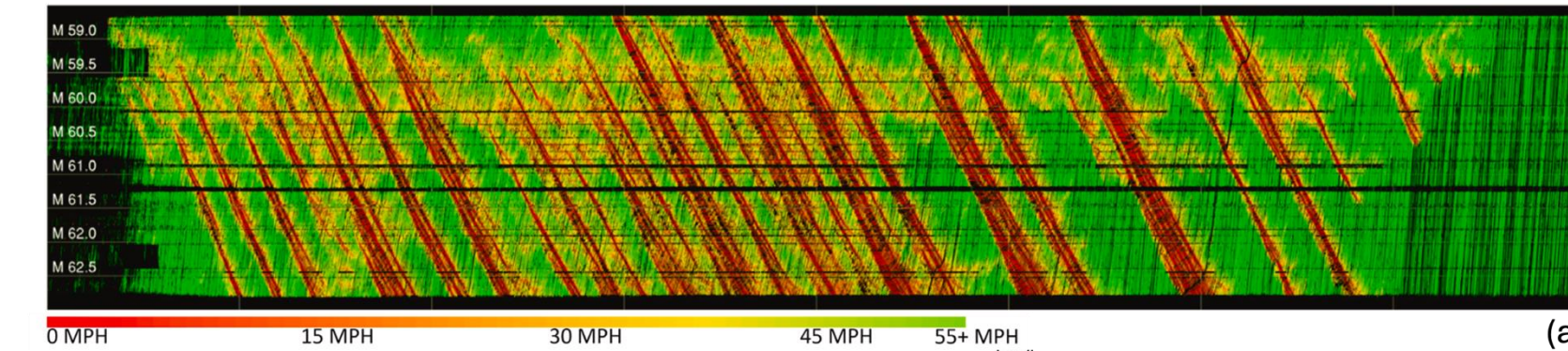
(b) Osaka City, Japan: 2 km

(c) California, USA: 0.6 km

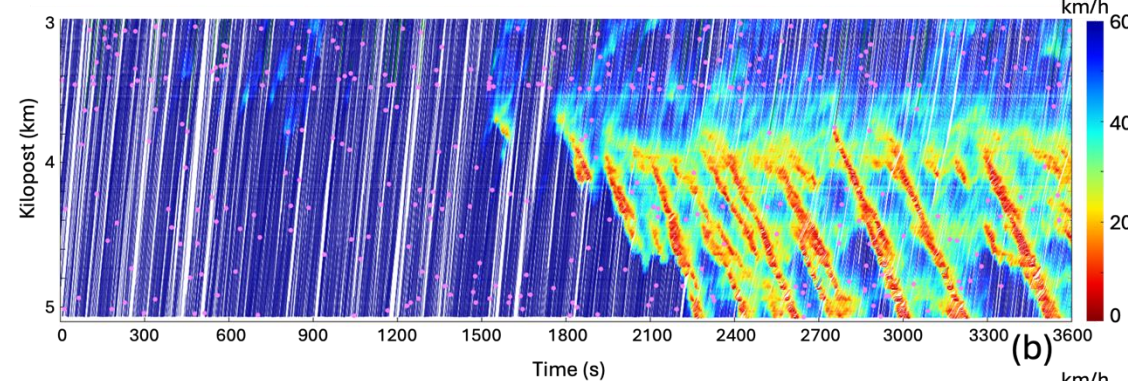
(d) Rotterdam, Netherlands: 23 km

(e) Florida, USA: 190 km

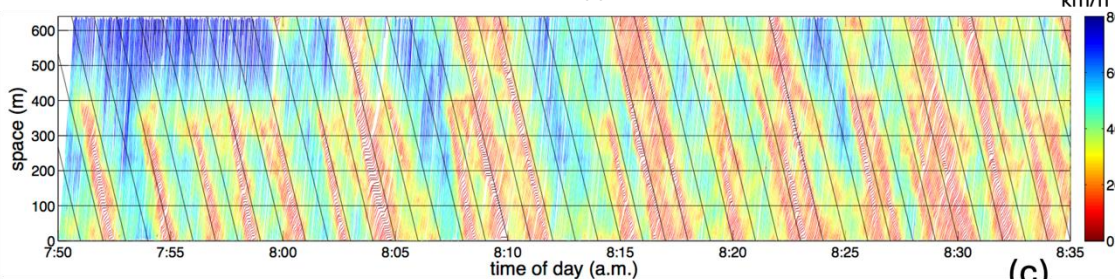
(f) London, UK: 18 km



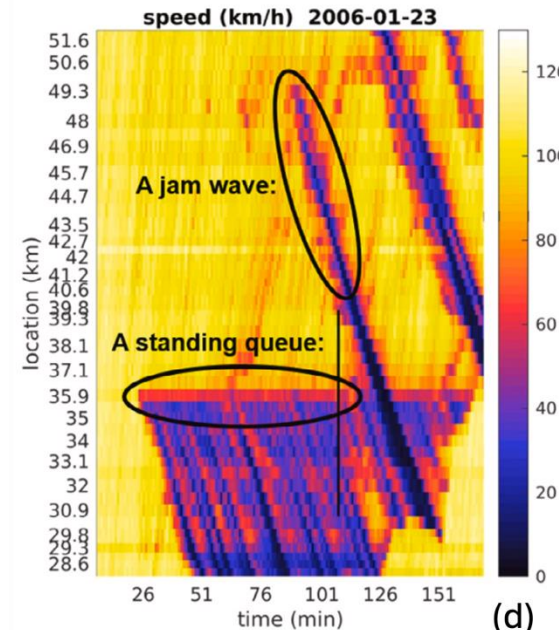
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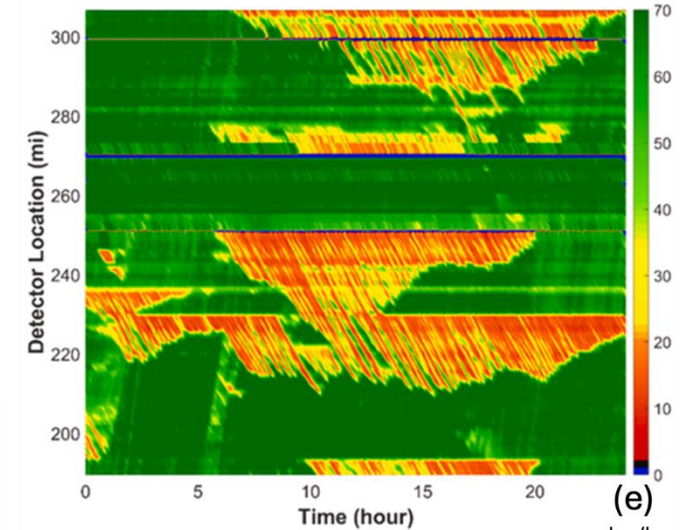
(b)



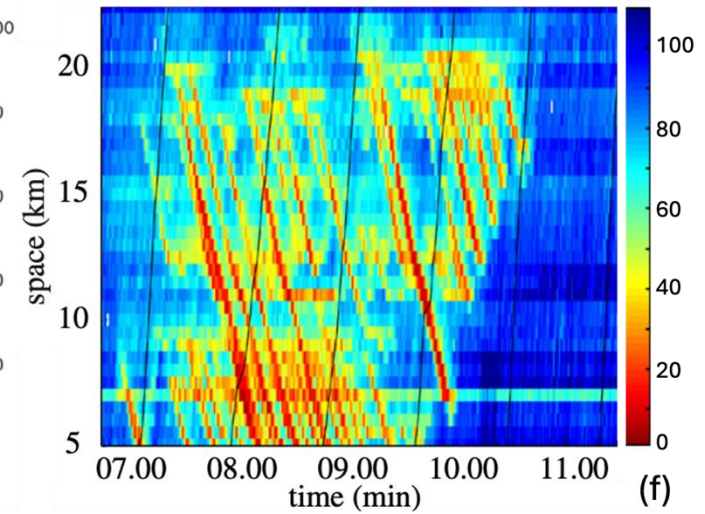
(c)



(d)

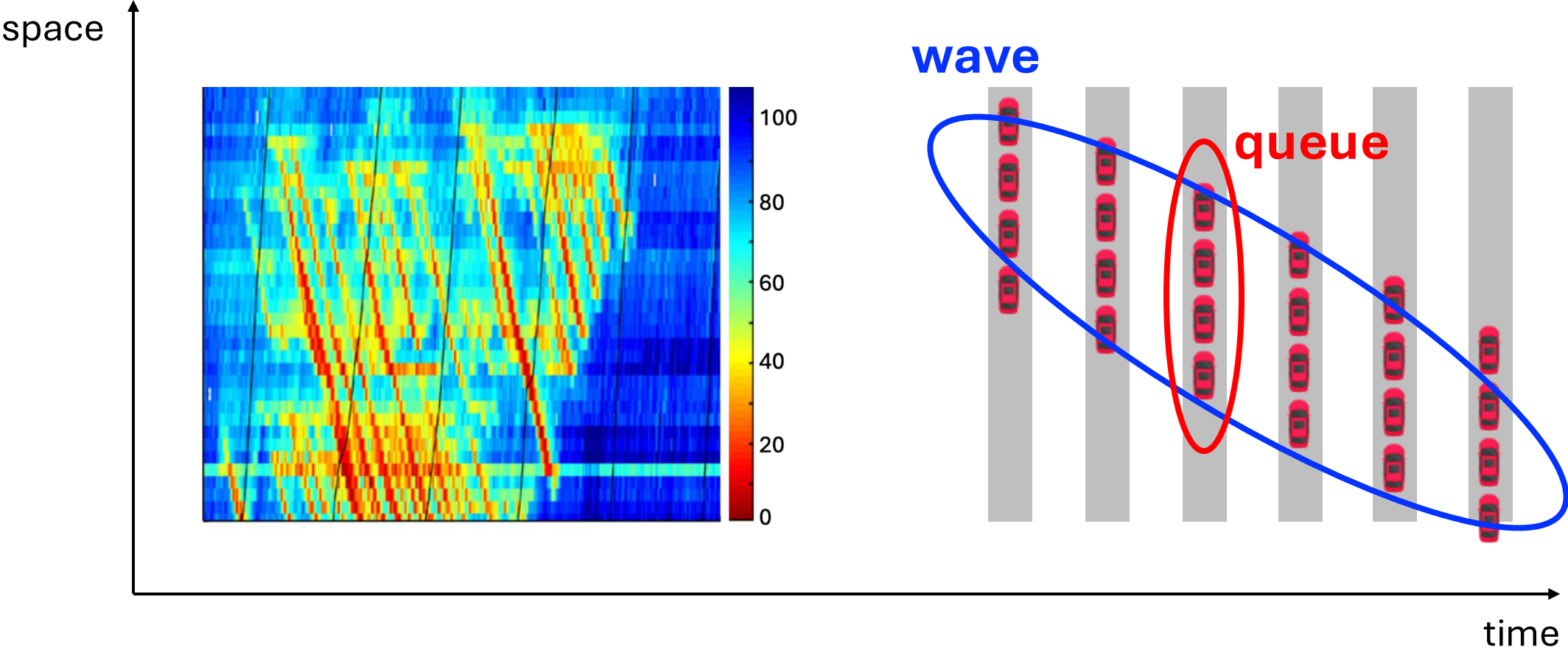


(e)



(f)

Stop-and-Go Waves on Freeways





moving queue

Stop-and-Go Waves on Freeways

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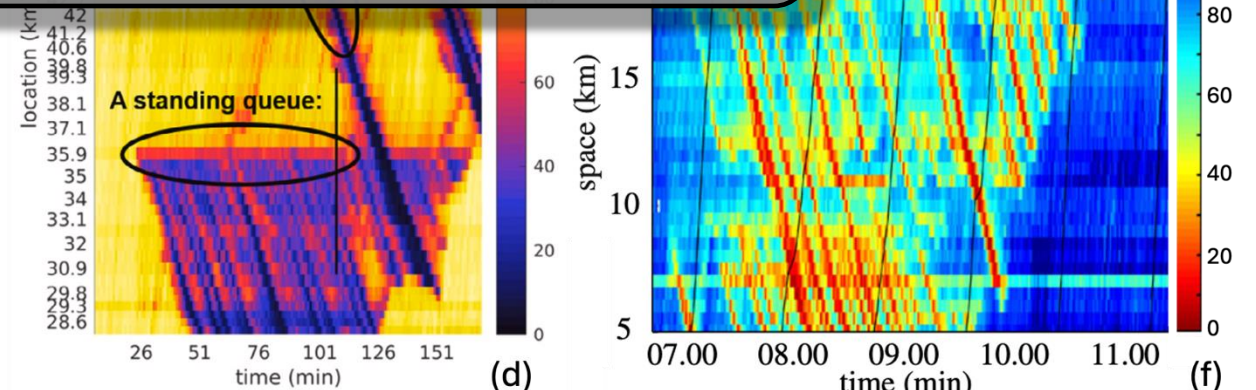
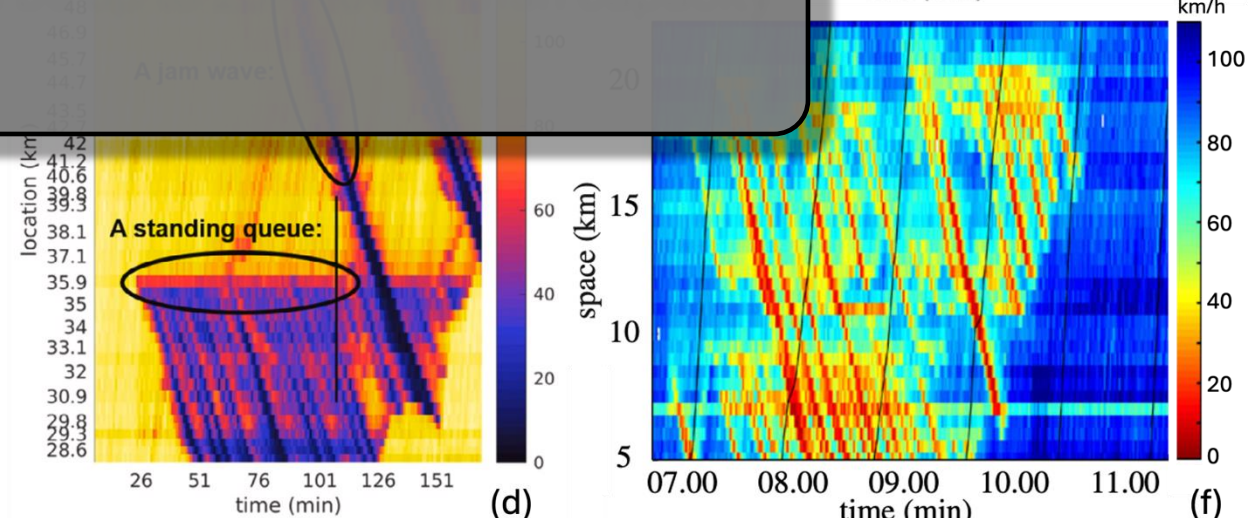
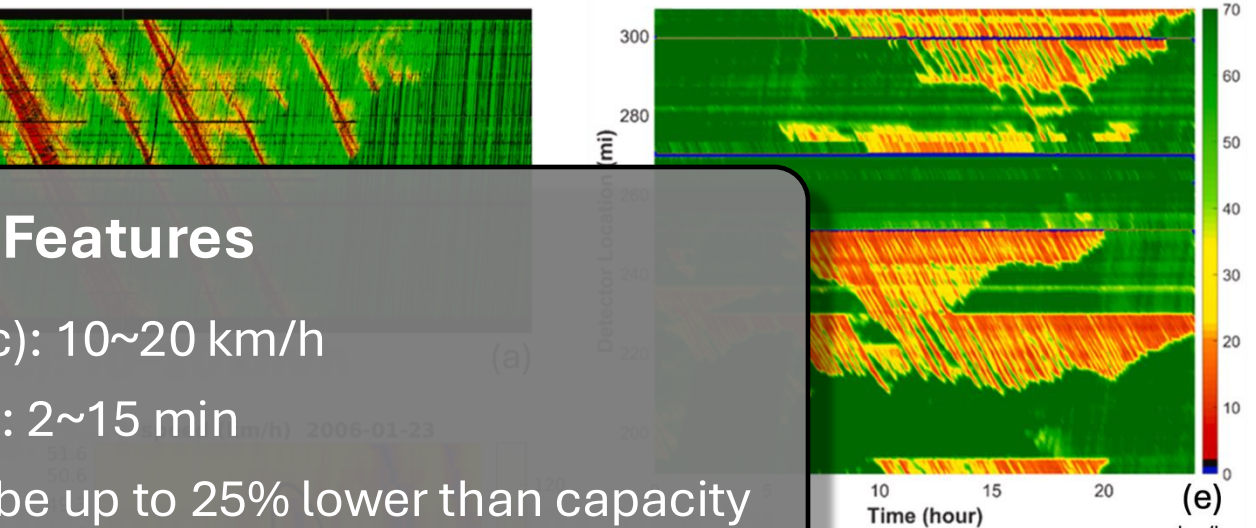
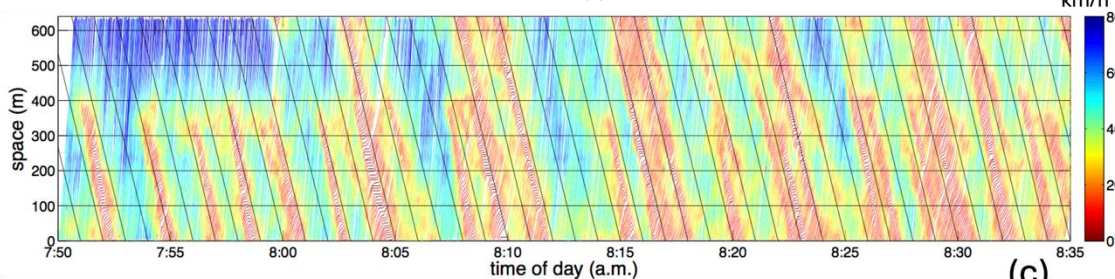
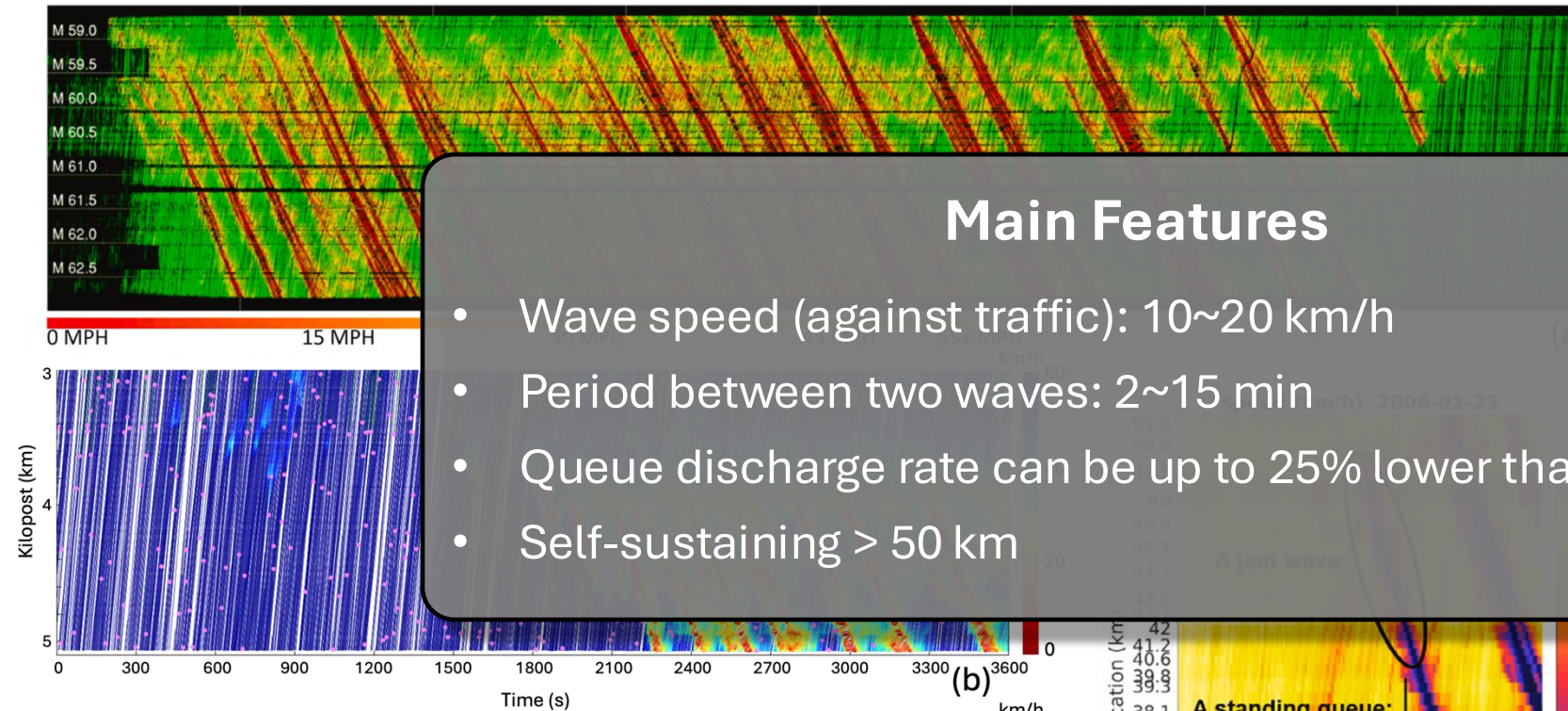
(d) Rotterdam, Netherlands: 23 km

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Main Features

- Wave speed (against traffic): 10~20 km/h
- Period between two waves: 2~15 min
- Queue discharge rate can be up to 25% lower than capacity
- Self-sustaining > 50 km

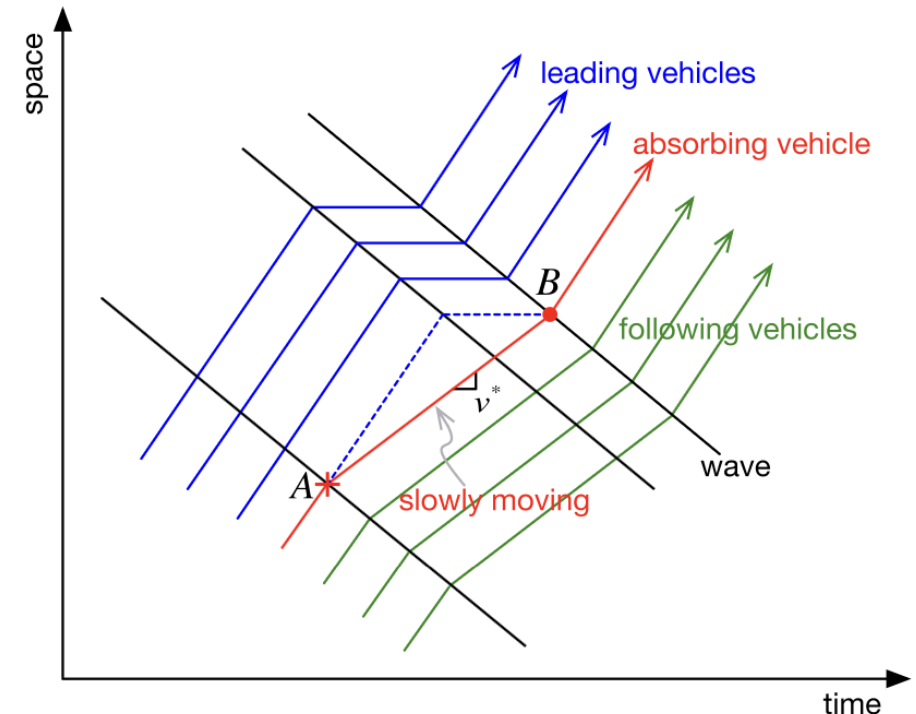


Stop-and-Go Wave Suppression: VSL vs JAD

VSLs are typically disseminated through VMS installed on gantries along the freeway. To ensure the effectiveness of an advanced VSL algorithm, **a series of gantries is required**, with the distance between two gantries commonly being about **500-1000 meters**, as seen in the Netherlands and the United States.



JAD is a driving strategy that guides a **specific vehicle to perform “slow-in” and “fast-out”** actions before being captured by a stop-and-go wave. Ideally, it can hold and **guide all vehicles behind to approach an upcoming stop-and-go wave** as planned, thereby achieving the goal of stop-and-go wave suppression.



Stop-and-Go Wave Suppression: VSL vs JAD

Similarities

Aspect	Details
Goal	Suppressing stop-and-go waves
Affected object	Traffic flow
Underlying logic	Decelerating inflow
Theoretical question	(1) Can stop-and-go waves be suppressed? (2) Will secondary waves be triggered?
Methodological question	(1) When to activate the strategy? (2) How to implement the strategy?

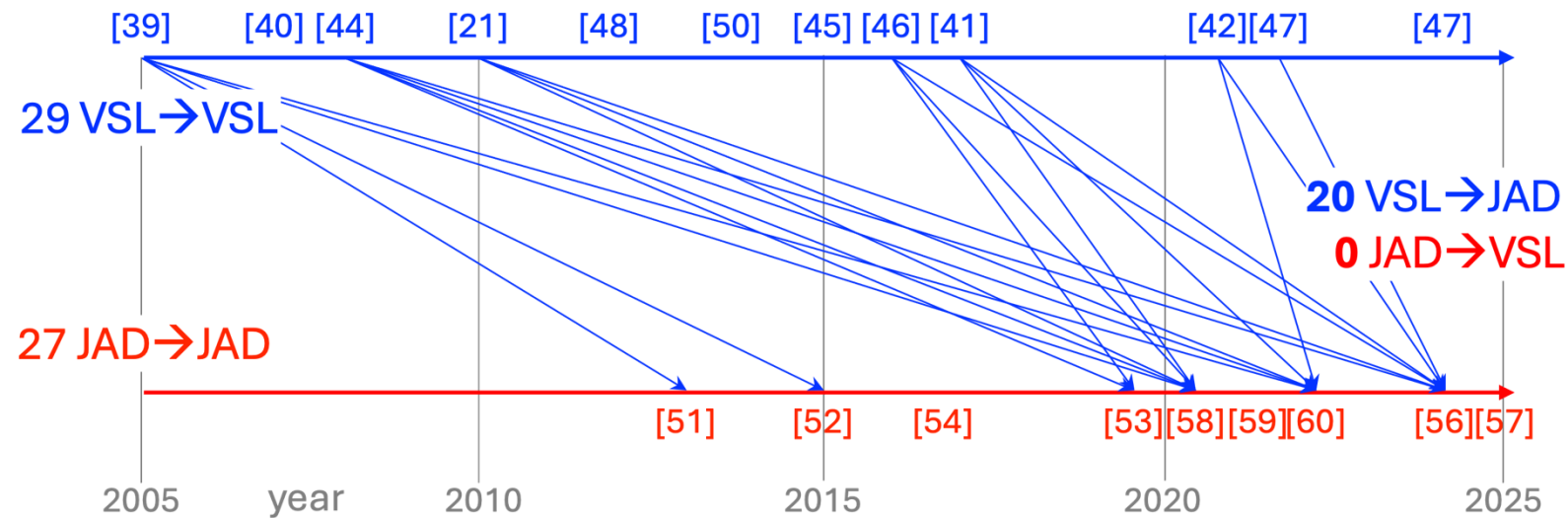
Differences

Aspect	Details	
	VSL	JAD
Main object of control	Speed limit	Vehicle
Feature of control object	Continuous in time, discrete in space, and value	Continuous in time, space, and value
Control source	Outside traffic	Inside traffic
Difficulty of execution	Low	High
Flexibility of execution	Low	High
Flexibility of deployment	Low	High
Cost of deployment	High	Low

Motivation: Disconnection between VSL and JAD

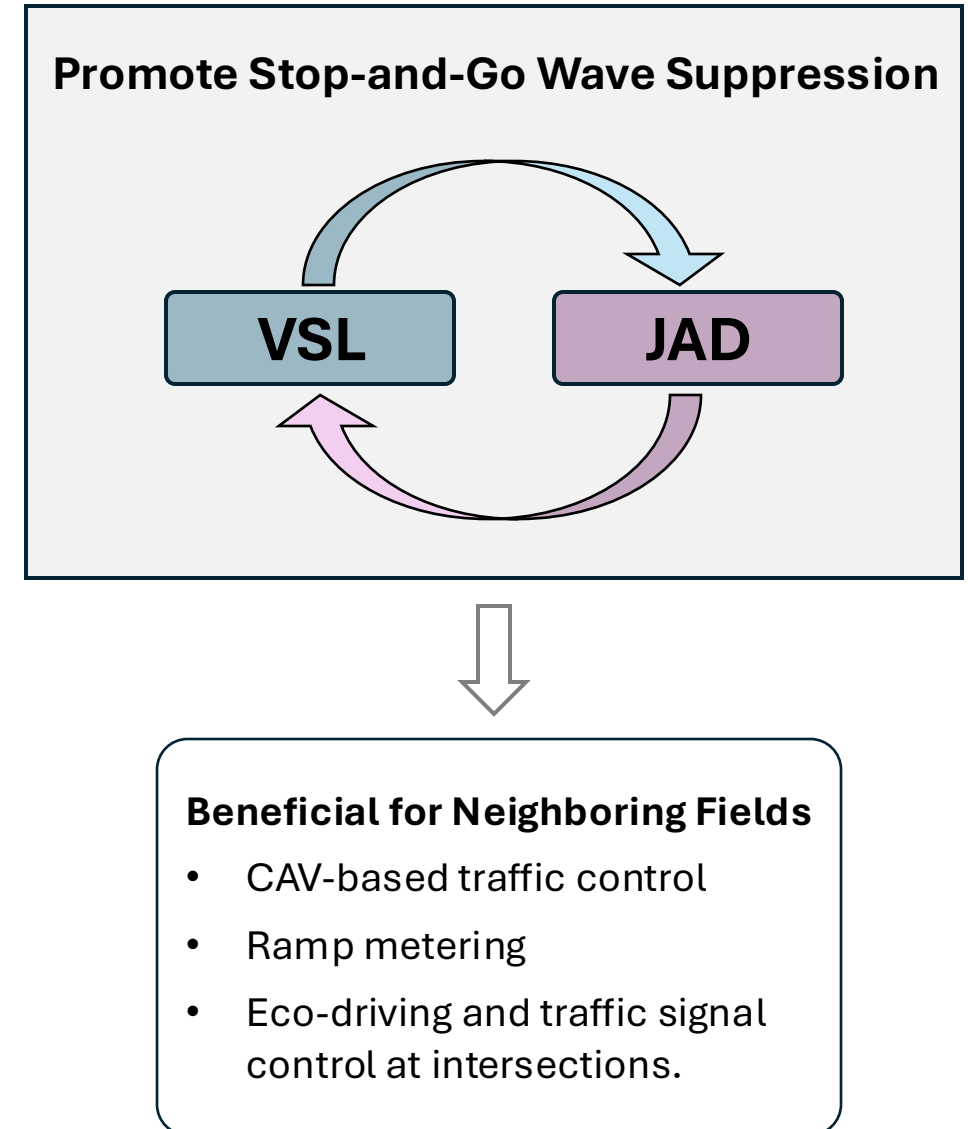
Two Observations

- Among the selected VSL and JAD papers, **none of the VSL papers have cited the JAD papers**, whereas the JAD papers tend to cite the VSL papers, although not comprehensively.
- Different terms, leading to difficulties in discovering each other's work through search engines
 - **VSL: Shock wave**
 - **JAD: Stop-and-go wave, oscillation**



Contribution

- This paper systematically reviews **the current progress in freeway stop-and-go wave suppression** through two techniques: VSL and JAD. By leveraging the strengths of one area, the other can effectively address its limitations, thus **promoting the overall research goal** of stop-and-go wave suppression.
- We carefully **identify important research gaps in both areas and highlight research opportunities**. In searching and reviewing related papers, we did find that some works were not developing along the sound direction. Therefore, we hope that this review not only inspires future work but also **helps avoid inappropriate assumptions and methods** in future studies.
- This review would also be **beneficial for researchers in neighboring fields**, such as CAV-based traffic control, ramp metering, and even eco-driving and traffic signal control at intersections.



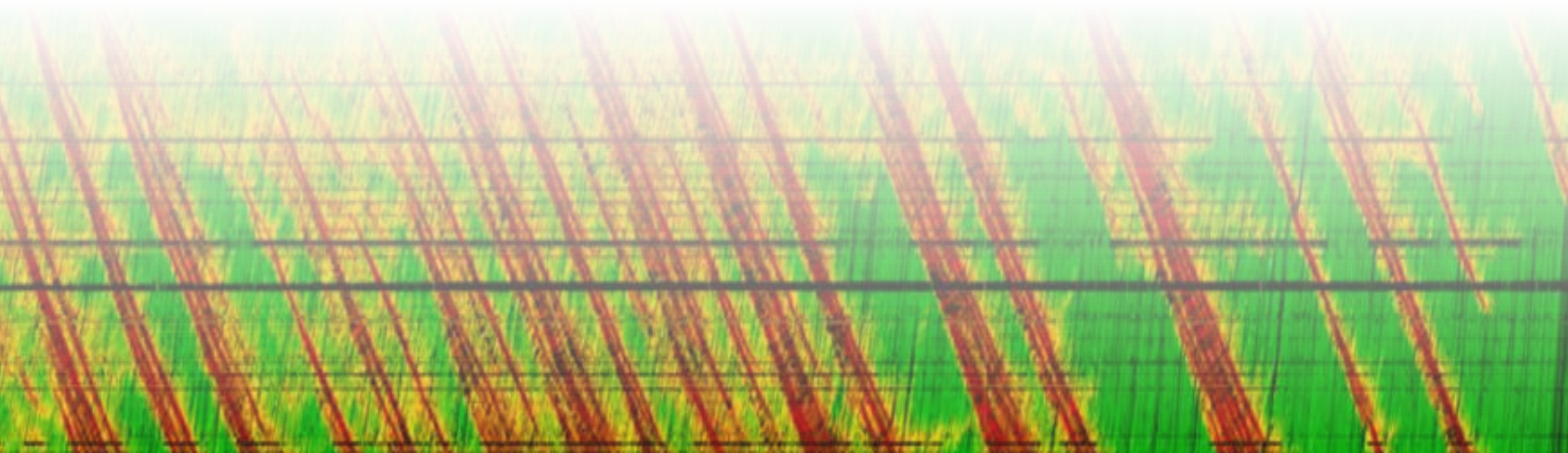
Scope

We focus solely on the stop-and-go wave suppression problem, in which **stop-and-go waves have already fully developed**. The following topics are excluded to make it a more specialized review paper

1. We do not discuss **speed homogenization-based congestion mitigation**.
 - A distinguishing difference between JAD and speed homogenization: the penetration rate of the controlled vehicles.
 - JAD is expected to resolve the stop-and-go wave with **only several controlled vehicles**
 - Speed homogenization commonly needs to control **more vehicles**
2. We exclude studies that aim to alleviate traffic jams **by addressing the bottlenecks** that trigger them, such as freeway ramps and lane drops.
3. We primarily focus on **works published after 2003**, as one may refer to the review paper {Papageorgiou2003} for earlier freeway control strategies.
 - before 2005, the stop-and-go wave suppression problem received little attention or research.

Summary of Stop-and-Go Wave Suppression Strategies

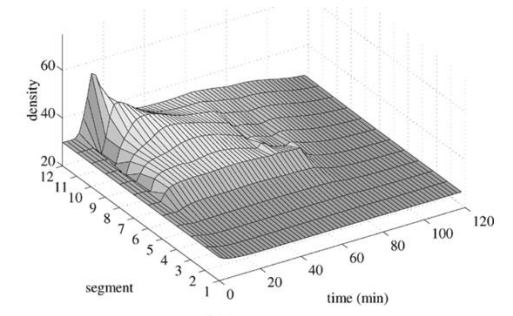
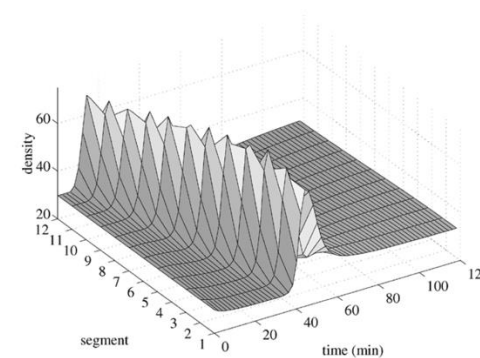
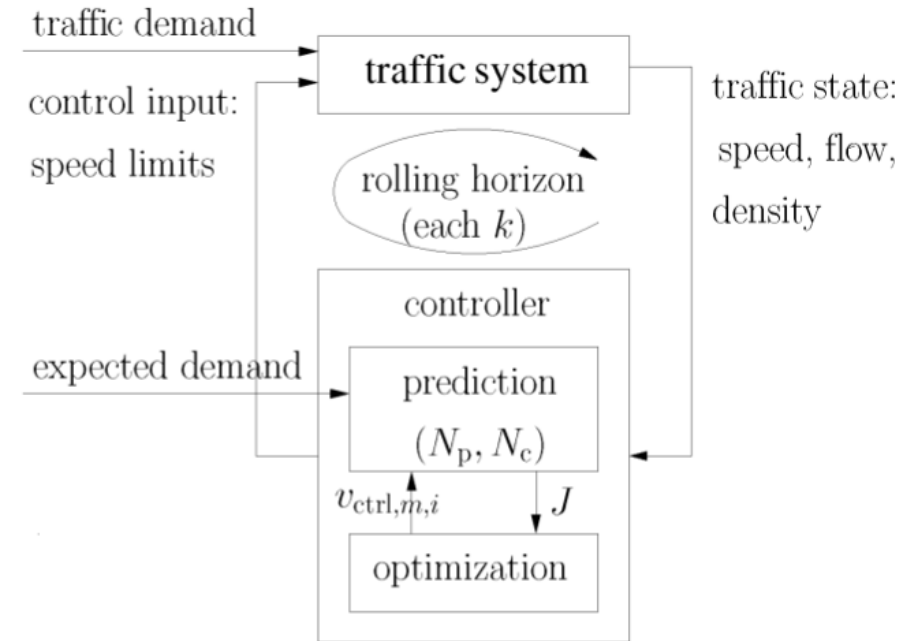
Variable Speed Limit vs. Jam-Absorption Driving



Variable Speed Limit

MPC-based

- In the early stage, the MPC was **built upon METANET** {VSL2005}, which is a second-order, deterministic, cell-based macroscopic traffic flow model.
- The strategies divide a freeway stretch into a series of cells (i.e., segments). Each cell, approximately **300 to 1000 meters in length, corresponds to a VSL**.
- The MPC strategy optimally **coordinates these VSLs** to **minimize total travel time**. It is known that the effectiveness of MPC is directly determined by the accuracy of the predictive model used, turning out that **METANET is the key to its effectiveness**.
- Test the METANET-based MPC solely within a METANET-based simulation, which is equivalent to assuming perfect prediction.



Variable Speed Limit

MPC-based

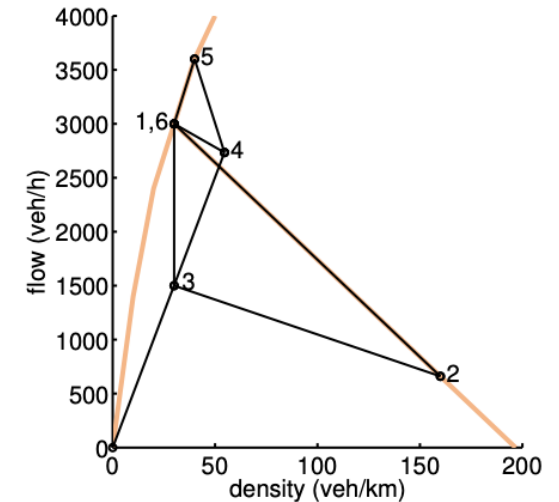
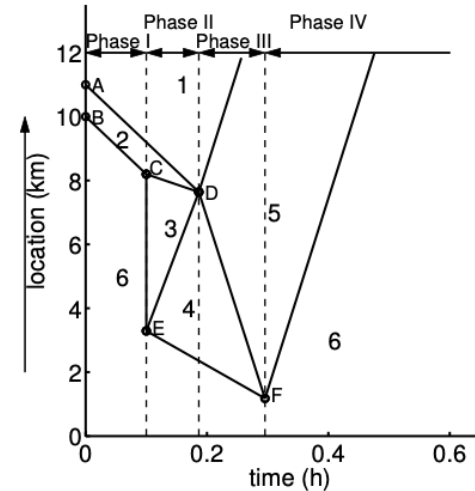
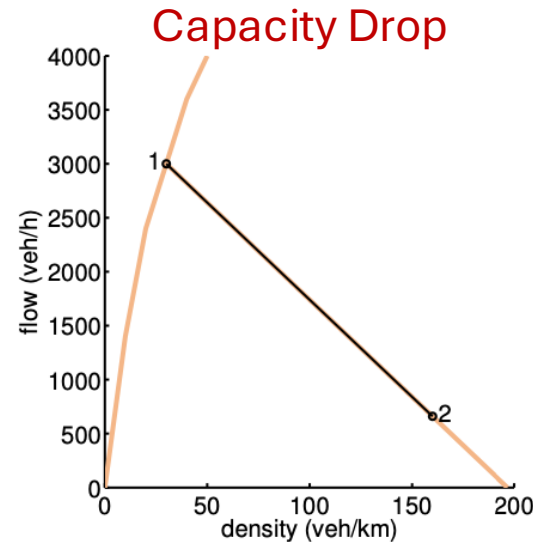
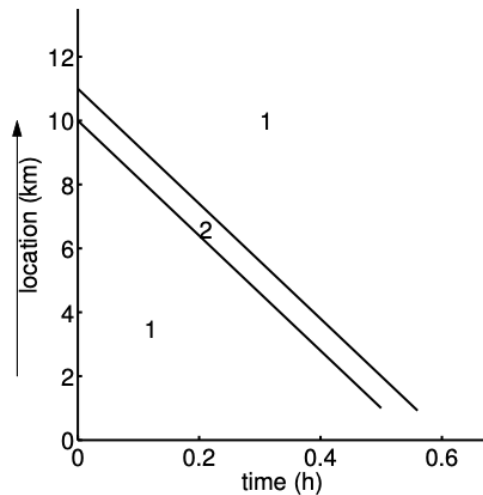
- To relax the assumption, {VSL2007} conducted a more rigorous and practical validation using the **microscopic traffic flow simulation** software Paramics 5.1.
- Further, to address the high computational burden associated with nonlinear and non-convex MPC, {VSL2017} proposed a **linear MPC-based strategy** with a **modified first-order CTM**, achieving similar optimization results with significantly improved efficiency.
- {VSL2021} introduced a **linear Lagrangian MPC**, which utilizes **average vehicle speeds as decision variables rather than flow**. This modification bridges traffic-oriented control and vehicle-oriented control, enhancing the flexibility of the overall framework of the stop-and-go wave suppression.

Type	Feature	Paper	Year	Modeling		Validation	
				Control Object	Details	Model	Waves
VSL	MPC	[45]	2005	Traffic	METANET-based prediction	METANET	Single
	MPC	[46]	2007	Traffic	METANET-based prediction	Paramics	Multiple
	MPC	[10]	2017	Traffic	Linear MPC; Modified CTM	METANET	Single
	MPC	[47]	2021	Traffic, <i>Vehicle</i>	Speed-based MPC with safety constraints. Upper: VSL; Lower: CAV	VISSIM, IDM+ [48]	Single

Variable Speed Limit

KWT-based

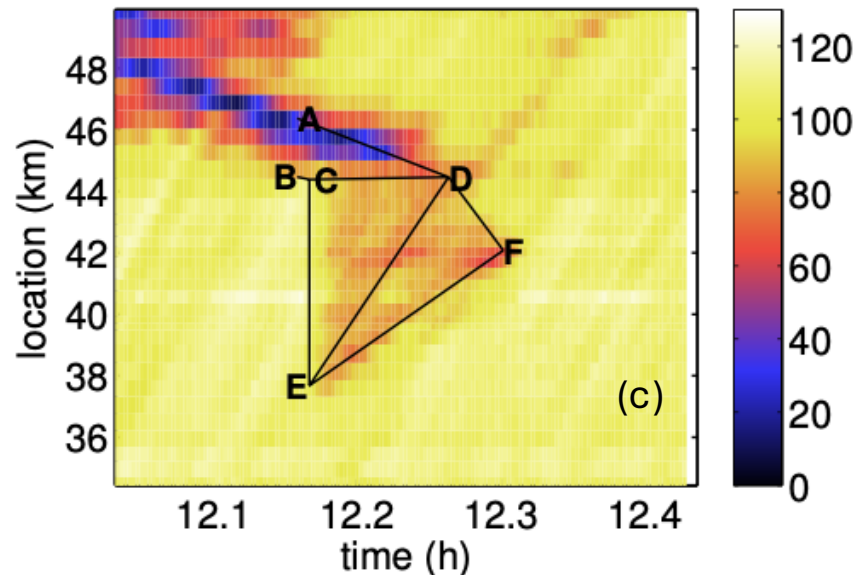
- {VSL2008} proposed the seminal algorithm (named **SPECIALIST**) by conducting a theoretical analysis using KWT on an FD with a capacity drop. **The opportunity to suppress a stop-and-go wave** arises from the empirical observation of an outflow reduction of **up to 30%** associated with a stop-and-go wave.



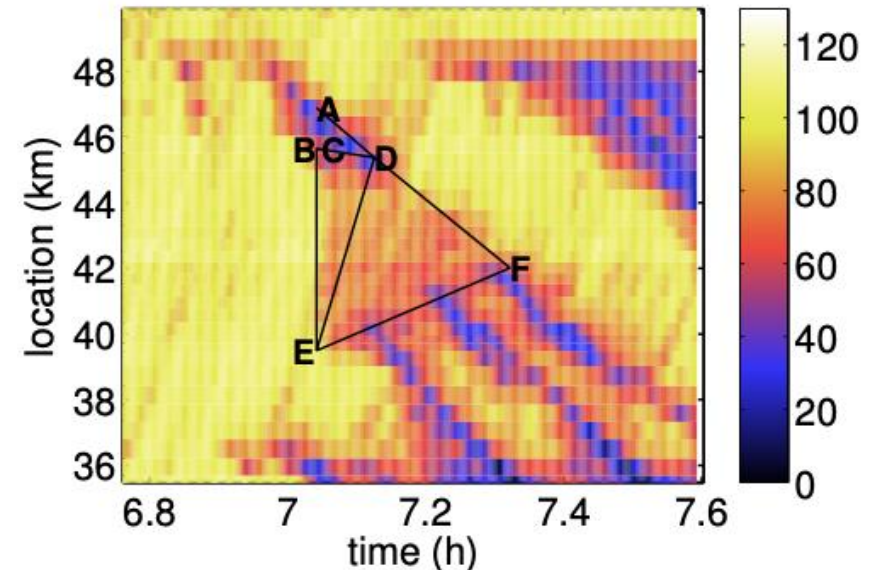
Variable Speed Limit

KWT-based

- In 2010, {VSL2010} conducted a **field test** of SPECIALIST to suppress stop-and-go waves over a **14 km stretch of the Dutch A12 freeway**, which, to date, may be the **only real-world test** of the stop-and-go wave suppression strategy.
- Although the effectiveness of the algorithm was well demonstrated, it was found that the algorithm was **frequently activated for other types of jams** rather than stop-and-go waves.
- It was reported that secondary waves were triggered by the newly imposed lower speed limit.



Real-world: **Success**



Real-world: **Second Waves**

Variable Speed Limit

KWT-based

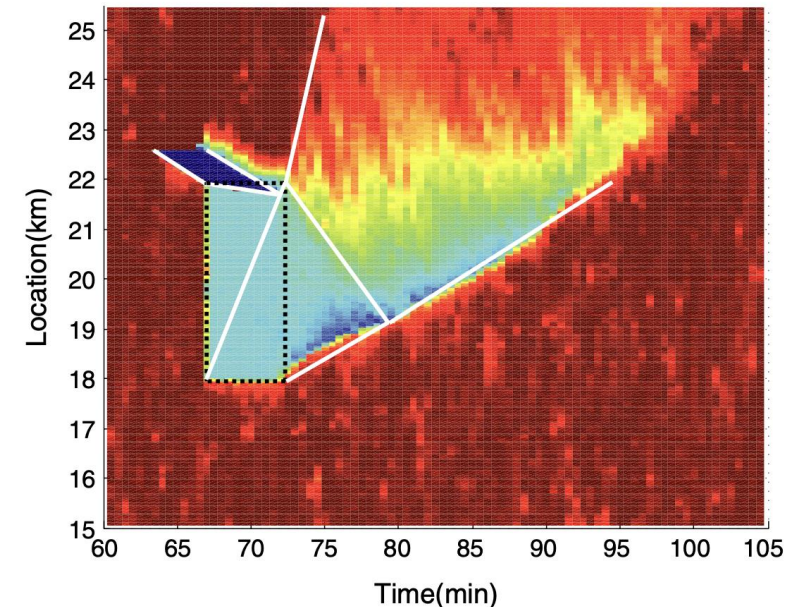
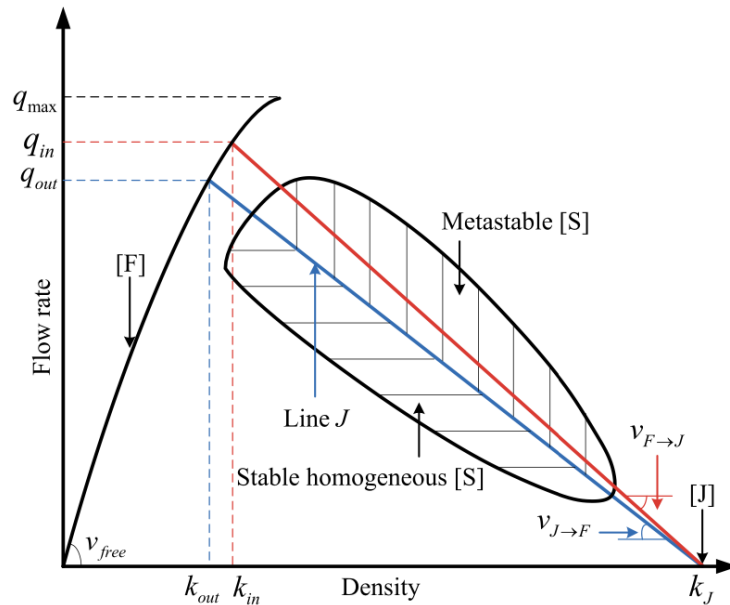
- {VSL2015} introduced ramp metering to the SPECIALIST framework and proposed a coordinated strategy with **mainstream control and ramp metering**.
- To take advantage of CAVs, {VSL2016} introduced a **bi-level framework to combine VSL and CAVs** and enhance the suppression.
- To fully accommodate connected vehicles, {VSL2024} converted the FD with a capacity drop into a TFD and proposed a Lagrangian strategy that can directly **broadcast VSL to connected vehicles**.

Type	Feature	Paper	Year	Modeling		Validation	
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	KWT	[9]	2008	Traffic	SPECIALIST (FD with capacity drop)	METANET	Single
	KWT	[28]	2010	Traffic	SPECIALIST	Real-world Test	Single
	KWT	[49]	2015	Traffic	SPECIALIST, Ramp metering	VISSIM	Single
	KWT	[50]	2016	Traffic, <i>Vehicle</i>	SPECIALIST, Upper: VSL; Lower: CAV	IDM+	Single
	KWT	[51]	2024	<i>Vehicle</i>	SPECIALIST (Triangular FD)	SUMO (3 lanes)	Single

Variable Speed Limit

TPTT-based

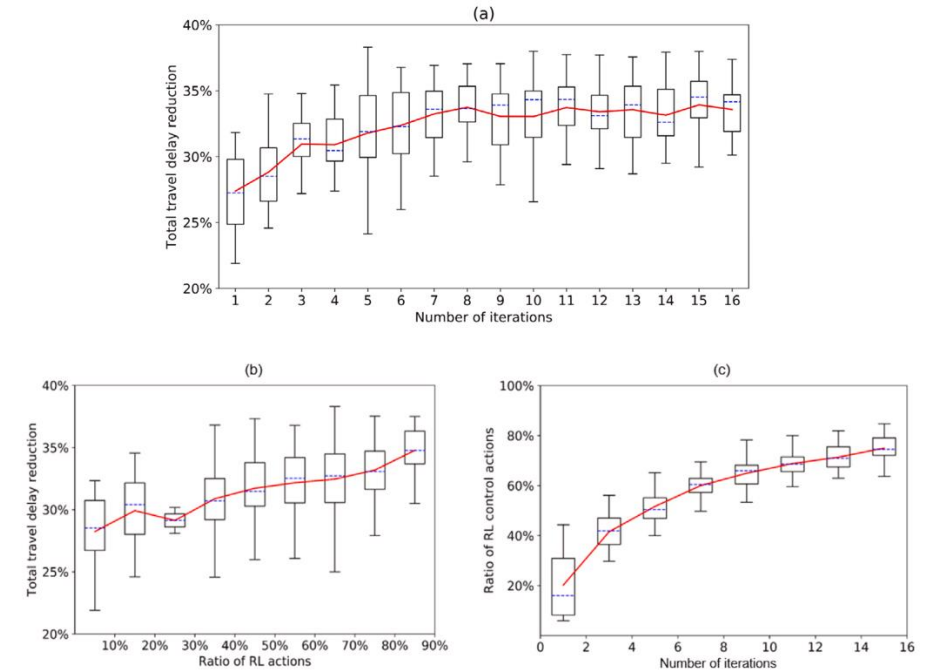
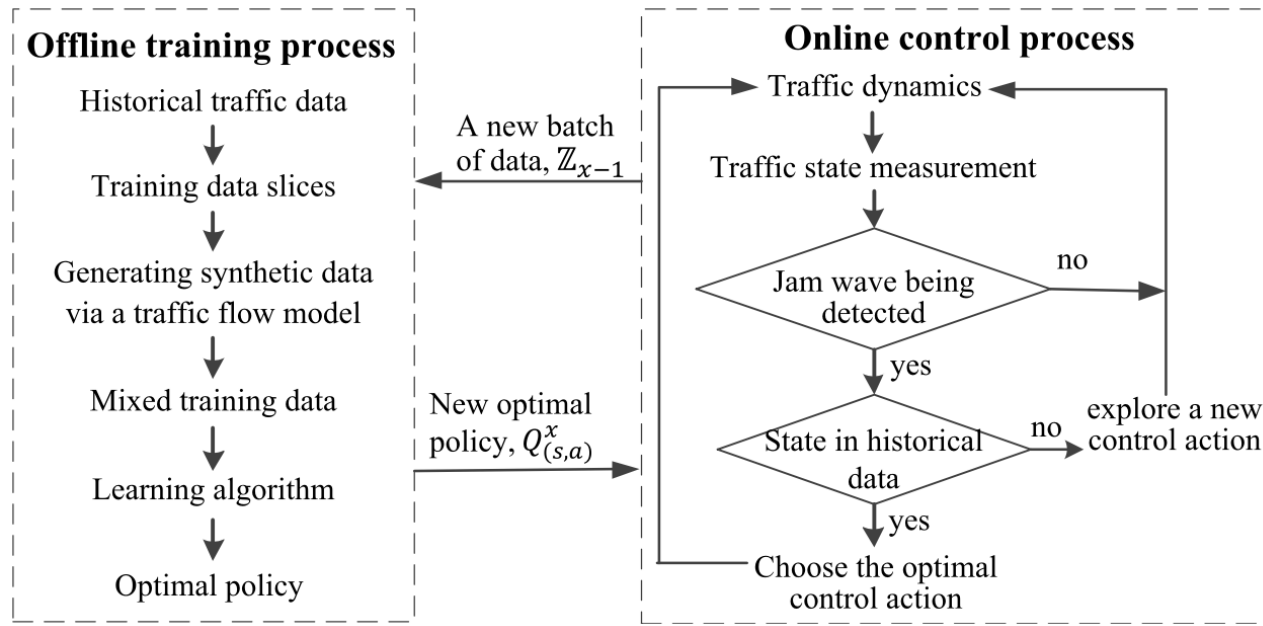
- As a theory emphasizing empirical observation, the **three-phase traffic theory (TPTT)** was also employed to investigate VSL strategies. As expected, the **new traffic flow model has led to new strategies and findings**, such as the traffic evolution different from the seminal one in {VSL2008}, although the strategy itself has not yet been empirically tested.



Variable Speed Limit

RL-based

- More recently, a model-free VSL strategy was proposed {VSL2022}. It employs an iterative training framework, where the optimal control policy is updated by **exploring new control actions both online and offline in each iteration**, thus avoiding the reliance on an RL model on a simulator.



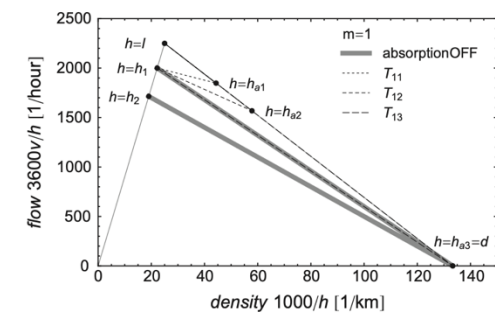
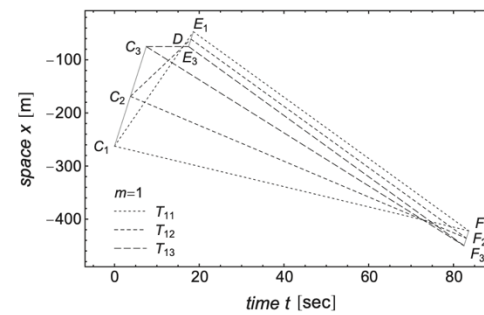
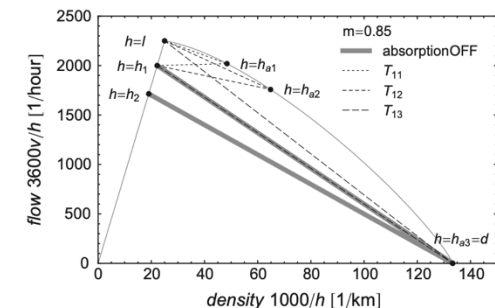
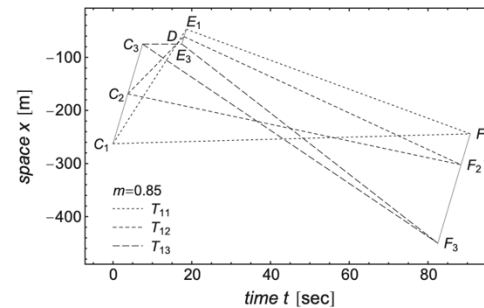
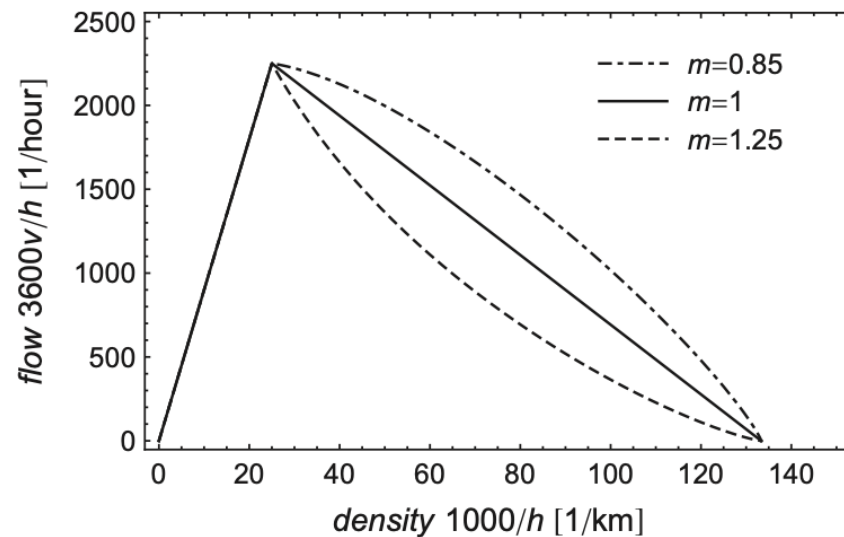
Variable Speed Limit: **Summary**

- A trend: **vehicle control**, as a supplement to traffic control, has attracted more attention recently
- Although **secondary waves** were observed in the real world as early as 2010, subsequent VSL studies have not paid much attention to this issue, which has been extensively discussed in JAD
- Most of the strategies were tested in a relatively simple scenario that included only **a single wave**
- In terms of adopting **AI technology**, it has been relatively slow.

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	TPTT	[52]	2012	Traffic	New traffic patterns found	KKS CA [53]	Single
	TPTT	[54]	2014	Traffic	Principle-based VSL coordination	KKS CA	Single
	RL	[51]	2012	Traffic	Model free: offline training + online control	METANET, Modified CTM [10]	Single

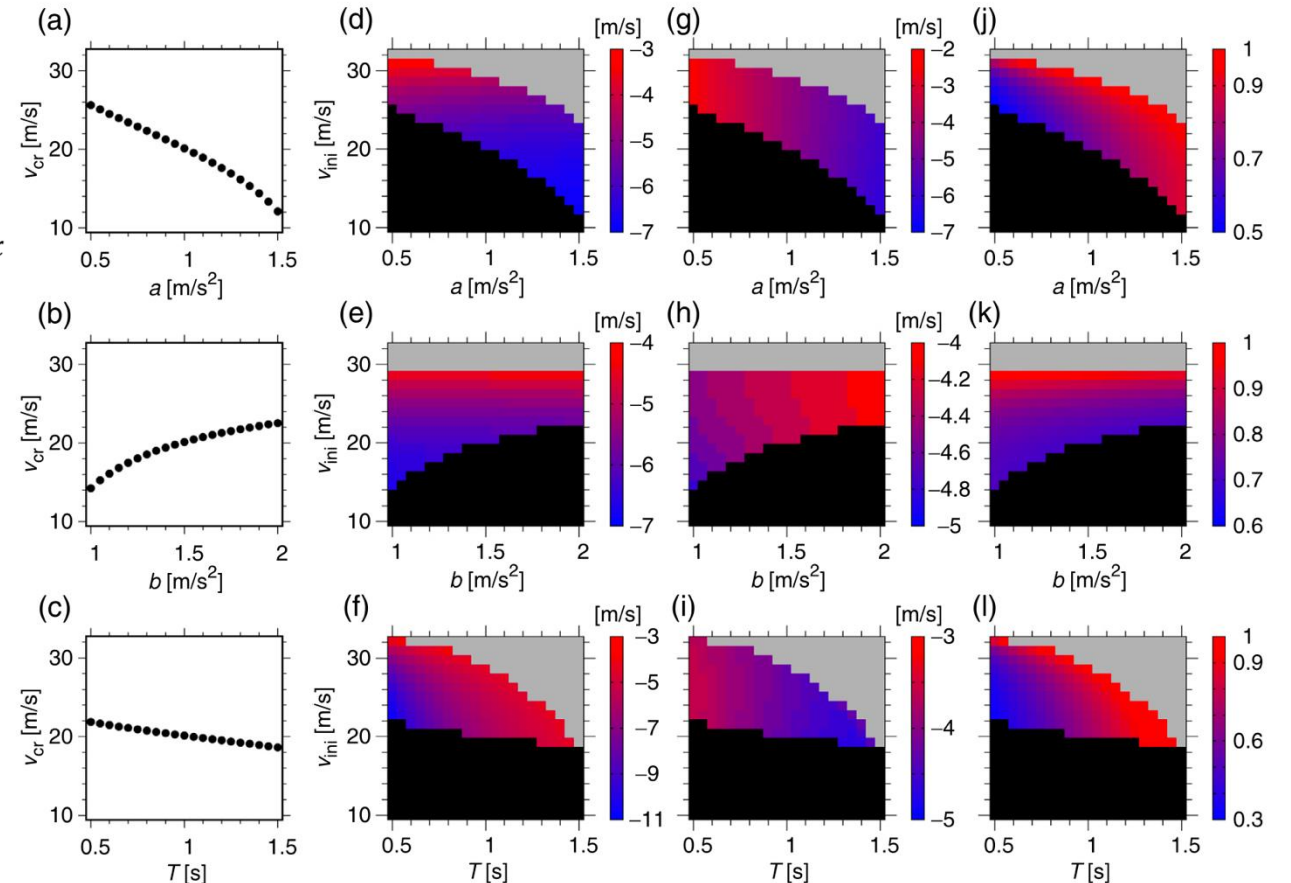
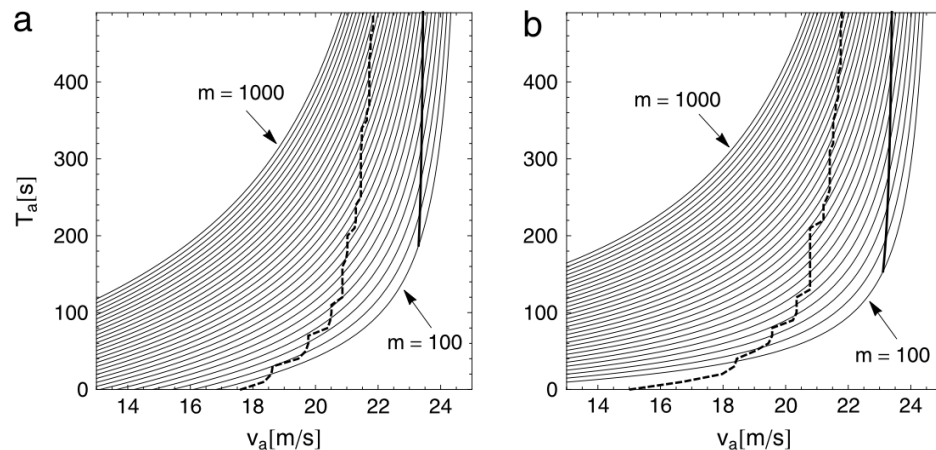
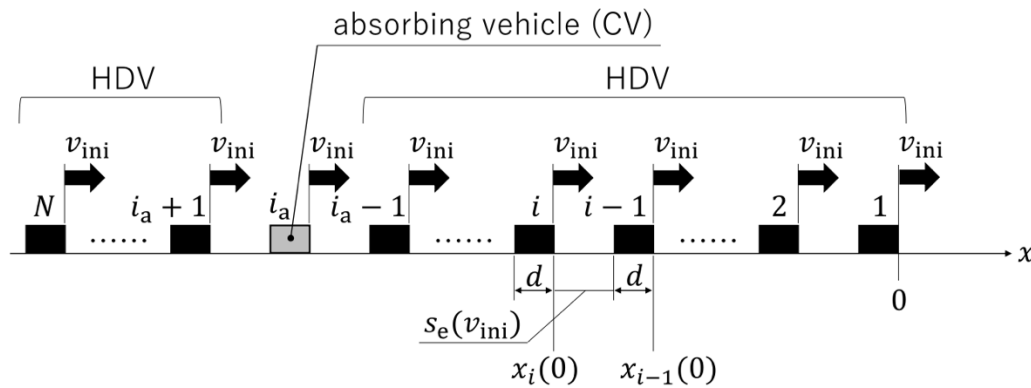
Jam-Absorption Driving

- The concept of JAD was initially introduced in a manner resembling a **popular science article**, as discussed by Beatty on a website. **It was not until 2013** that {JAD2013} sought to systematically examine and analyze its potential in a scientifically rigorous manner.
- From the very beginning, the research questions were precisely framed around **two key questions**: (1) Can JAD suppress stop-and-go waves? (2) Will secondary waves be triggered?
- Centered around these two questions, {JAD2013} conducted a theoretical analysis based on KWT, **focusing more on the impact of FD shapes** (i.e., concave, convex, and triangular FDs) rather than on a capacity drop



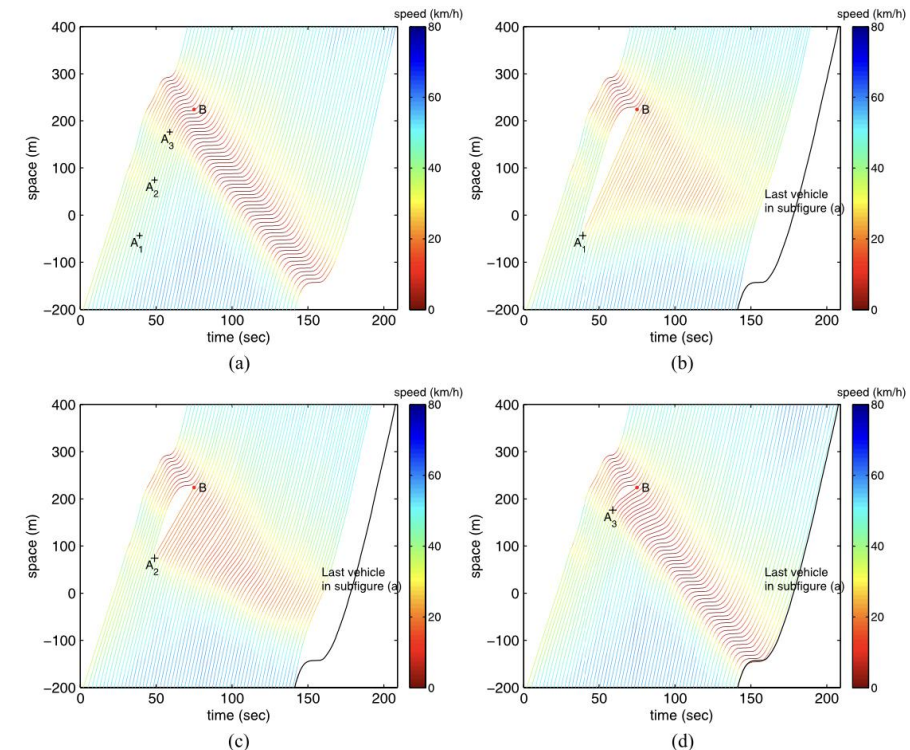
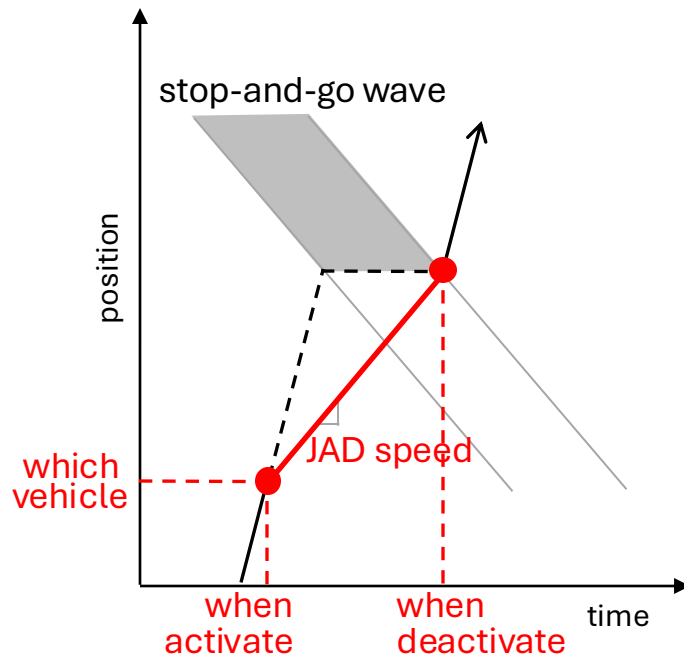
Jam-Absorption Driving

- Later, {JAD2015} and {JAD2020b} extended the analysis from a **microscopic perspective**. The former employed the Helly CF model and the IDM, and the latter the IDM. **The explicit formulations of vehicle dynamics in these CF models make the analyses traceable**. However, it is important to be aware that the conclusions rely on the strong assumptions inherent in these models.



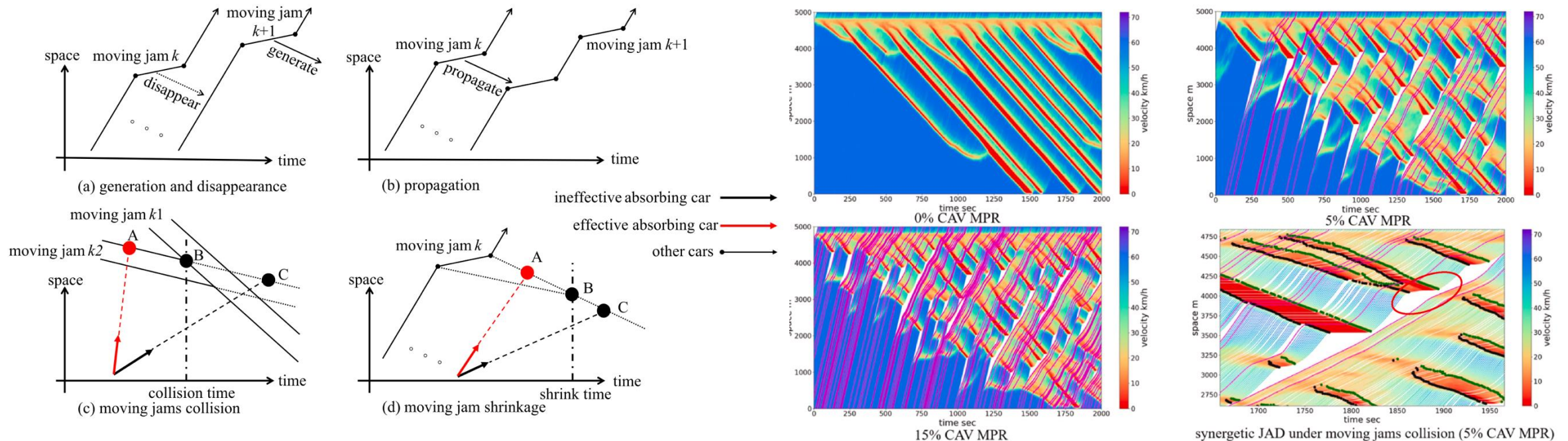
Jam-Absorption Driving

- Since {JAD2017} in 2017, **more practical implementation-oriented** JAD strategies were proposed. Given a predicted stop-and-go wave propagation track, **the core tasks are** to determine
 - When to activate and deactivate a JAD vehicle?
 - Which vehicle is appropriate to implement JAD?
- {JAD2017} **predicted the propagation** of a stop-and-go wave using the Newell CF theory and determined the JAD vehicle based on the principle of activating a JAD vehicle **as early as possible**.

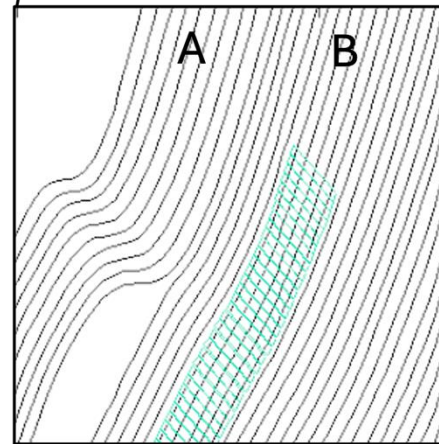
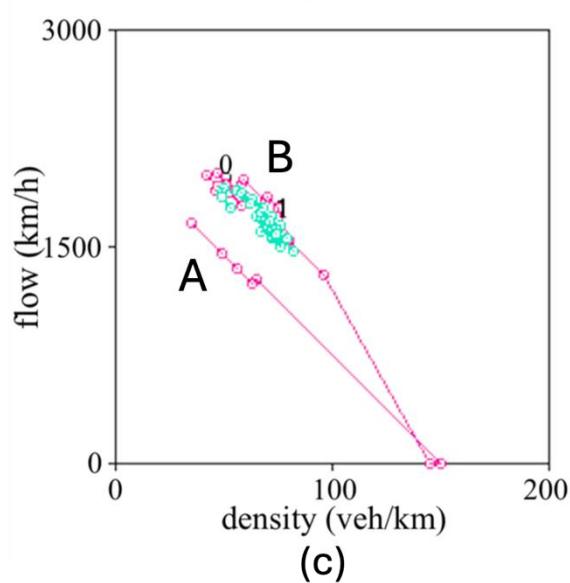
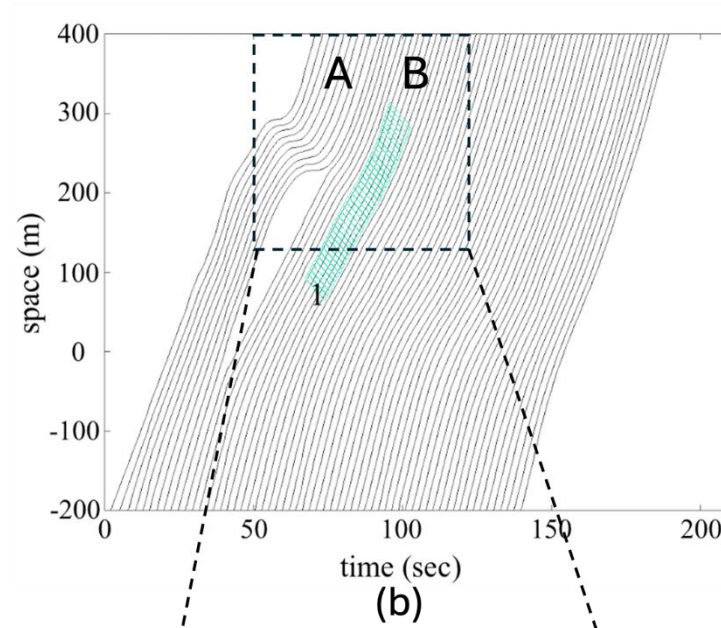
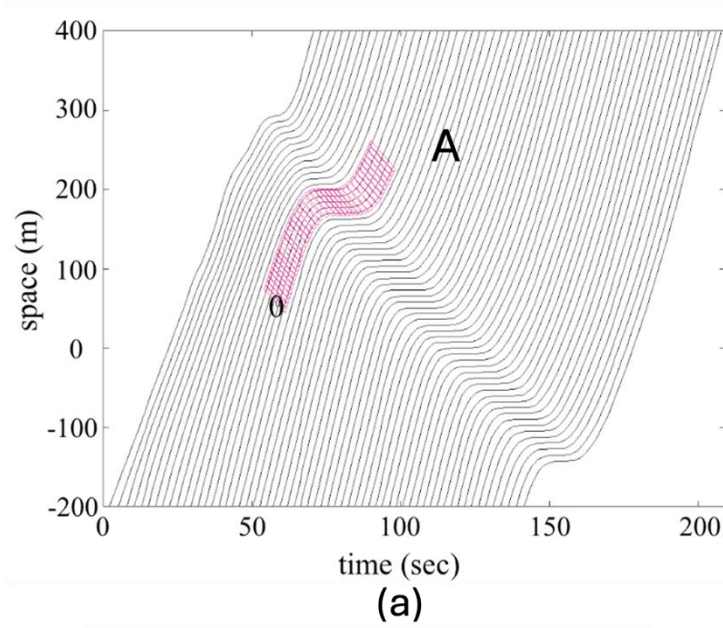


Jam-Absorption Driving

- Recently, {JAD2021b} introduced the concept of **a capacity drop** to the JAD study and proposed an optimal control-based JAD strategy that jointly **optimizes traffic efficiency and safety** by controlling the speed of JAD vehicles.
- {JAD2024b} targeted **multiple wave suppression** in one shot and designed a comprehensive set of steps to suppress the diverse evolutions of multiple stop-and-go waves.
- The latest work {JAD2025} **randomized the parameters** of Newell CF model to estimate the stop-and-go wave propagation and incorporated the JAD task into an **MPC-based vehicle control framework**.



Jam-Absorption Driving: **Summary**



- **JAD Opportunity**

Compared to **the queue discharge rate at A**, JAD results in a **higher flow at B**, even though the speed at B **cannot exceed** that at A due to the constraint imposed by the vehicles at A.

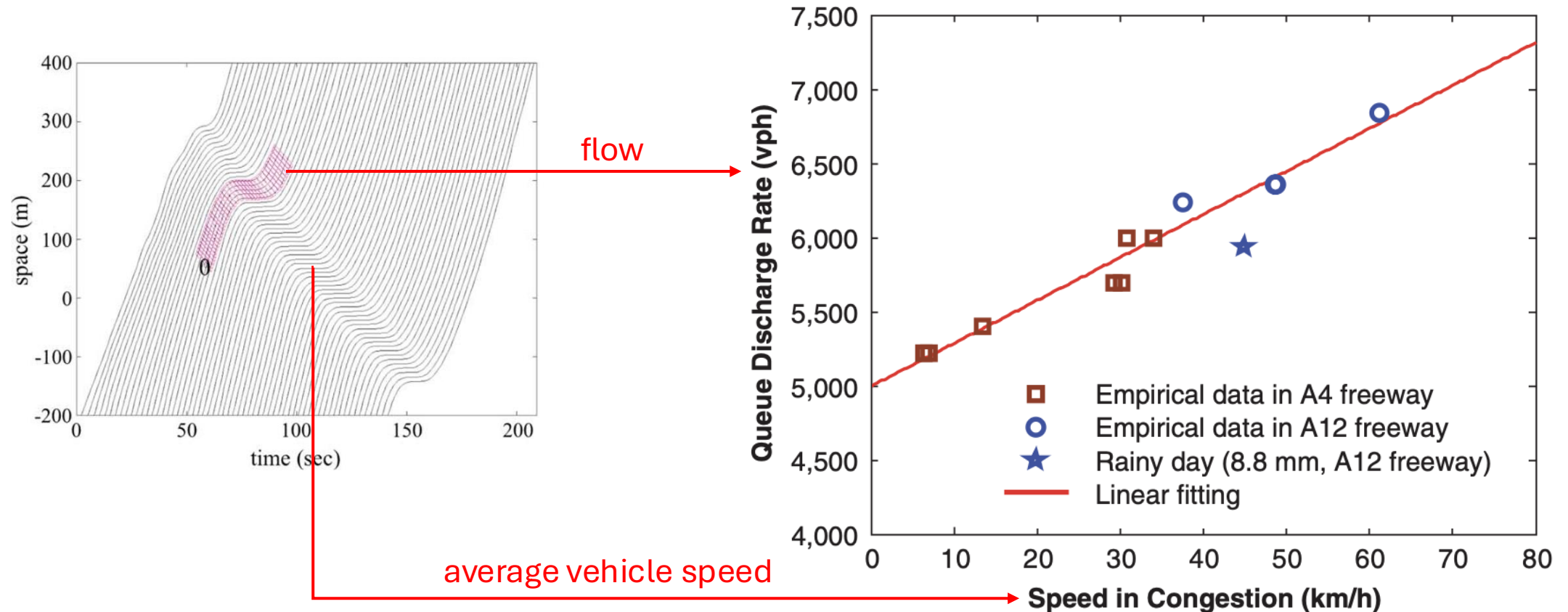
It should also be noted that the resulting high flow may be **unstable** and could lead to further breakdowns.

Jam-Absorption Driving: **Summary**

- Up until now, the issue that **secondary waves** could be triggered remains unresolved.
- The newly-triggered secondary waves may largely contribute to **increased travel time** when absorbing dense stop-and-go waves.
- Another reason related to secondary waves might be **the distance of JAD execution**
 - To effectively absorb a stop-and-go wave, JAD should be implemented over **at least several hundred meters** to ensure that the speed reduction from free-flow speed to JAD speed is gradual and less abrupt.
 - Controlled segments in current VSL strategies are **typically several kilometers long**, sometimes exceeding even ten kilometers, implying that the current distance of JAD execution may still be conservative, and the execution distance could be further extended.

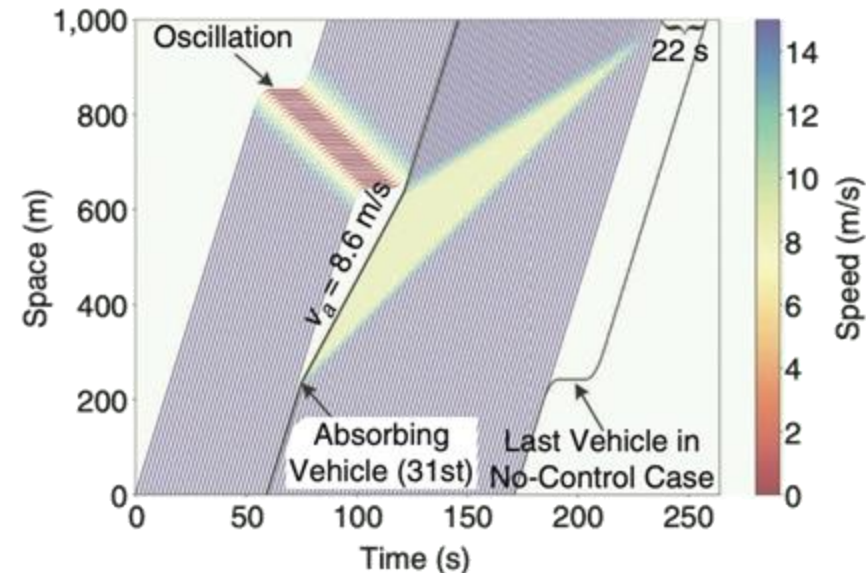
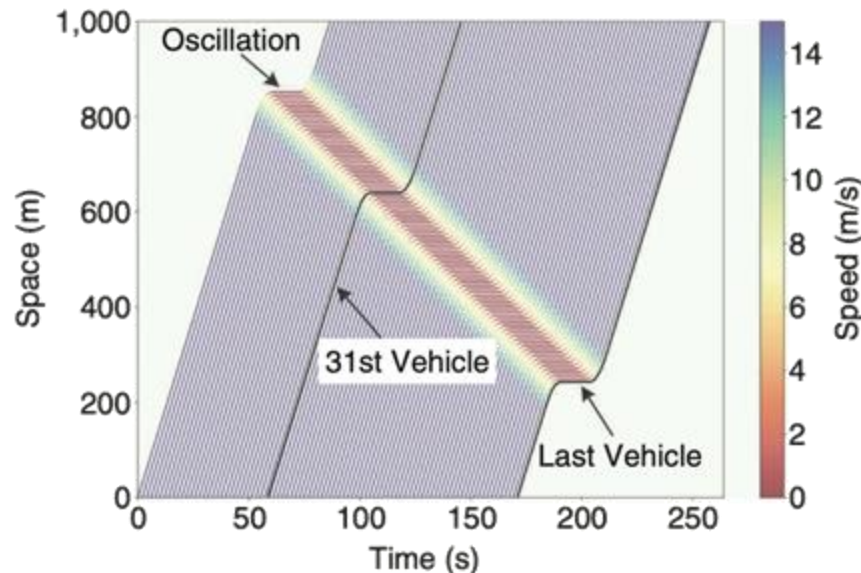
Jam-Absorption Driving: **Summary**

- As shown in {Yuan2014}, the queue discharge rate reduction is **positively and linearly related to the average speed of vehicles within a moving queue**, highlighting the potential benefit of suppressing a wave, especially when the speed within the queue is low.



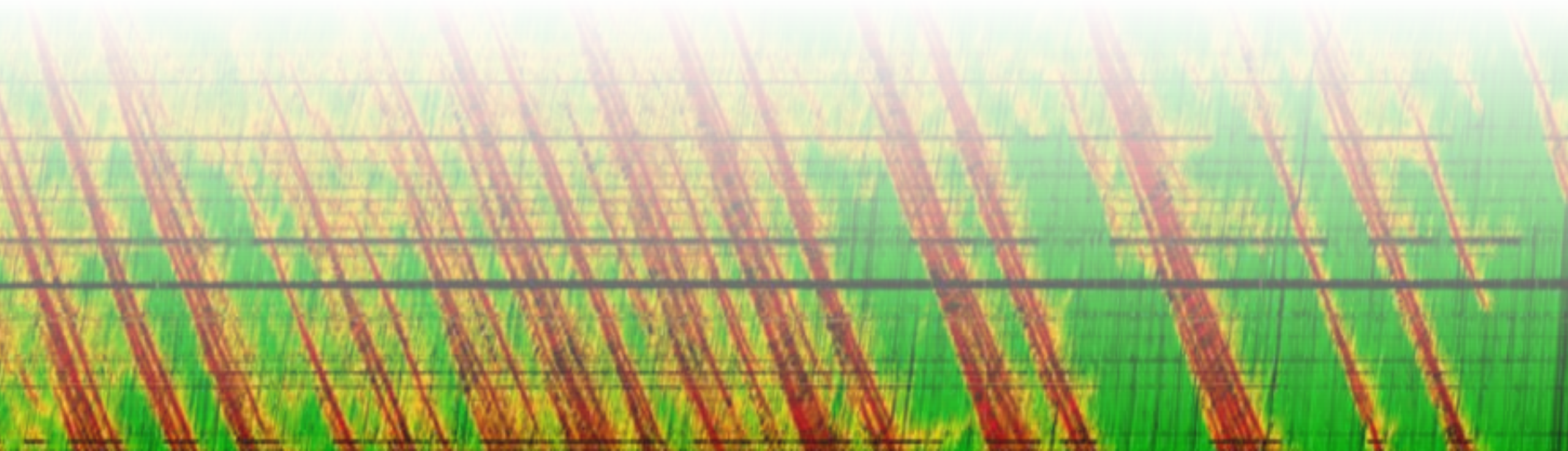
Jam-Absorption Driving

- A **growing trend in which a platoon of vehicles** has been employed and controlled to implement JAD for suppressing stop-and-go waves.
- However, this approach seems to **deviate from the original motivation behind the JAD strategy**, which was to suppress stop-and-go waves using a single (or very few) vehicle(s).
- We should be clearly aware that if we have the capability to stably and precisely control a vehicle platoon, the platoon-control-based JAD strategies should be **compared with high CAV penetration-based speed homogenization strategies**, rather than with the single-vehicle-based JAD strategies.



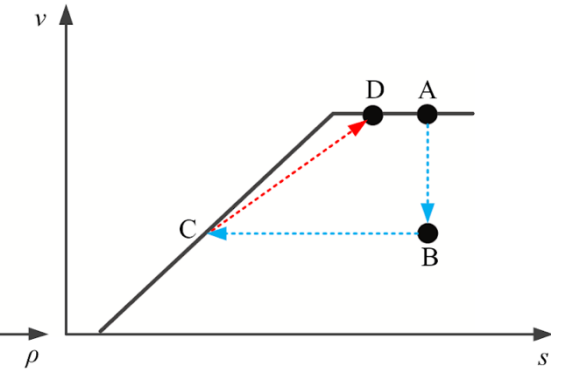
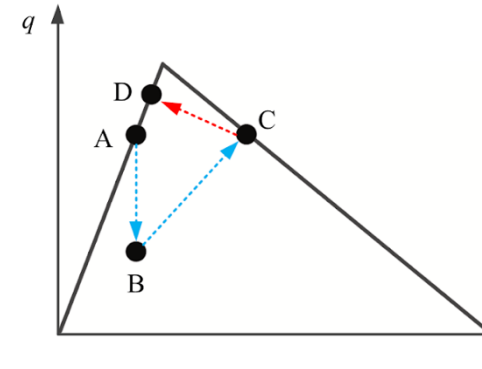
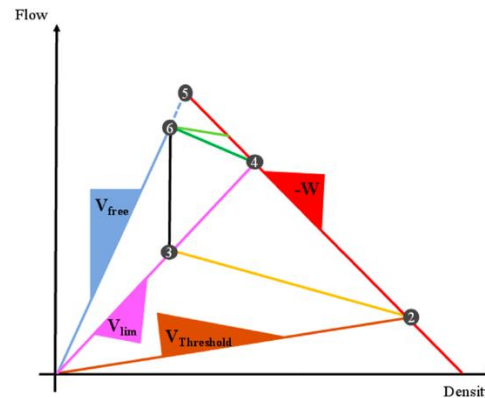
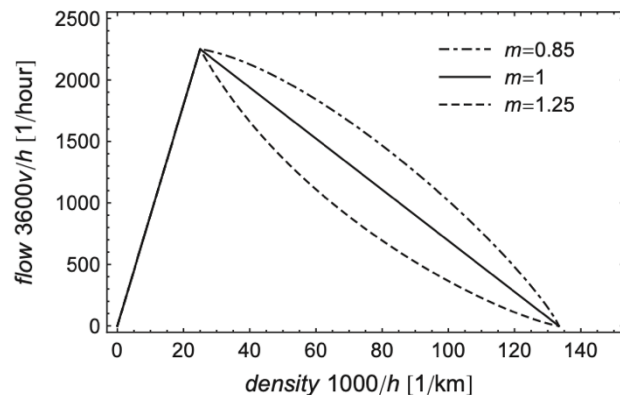
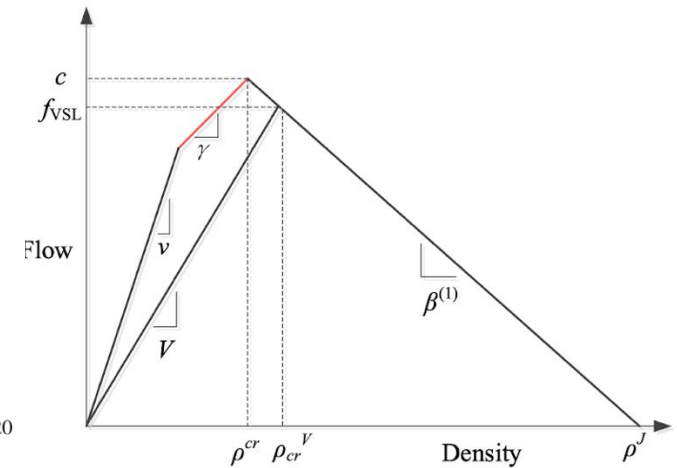
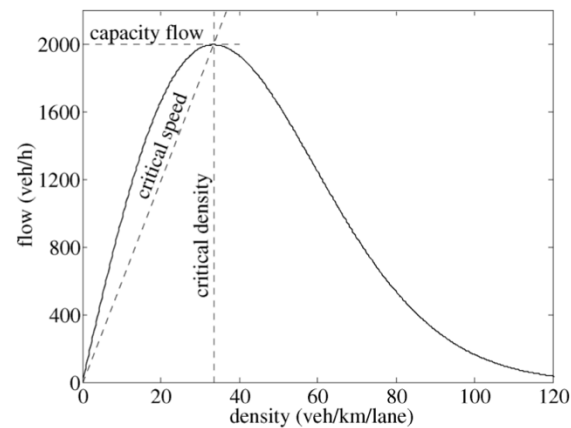
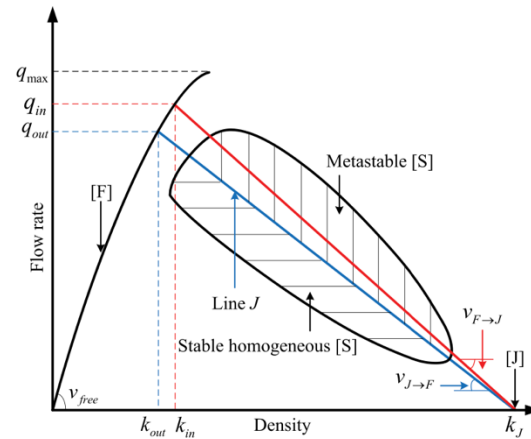
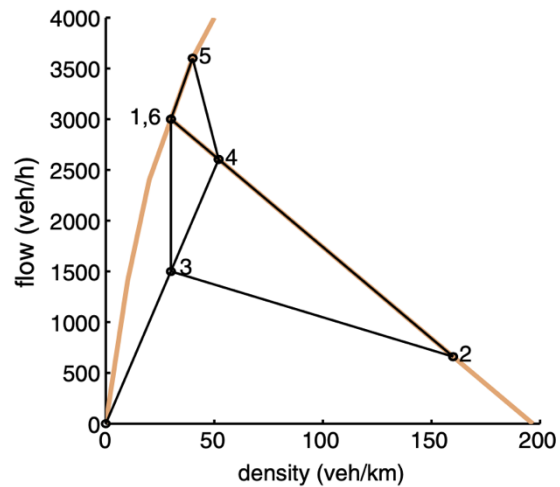
Bridging and Research Opportunities

Variable Speed Limit vs. Jam-Absorption Driving



Fundamental Diagrams

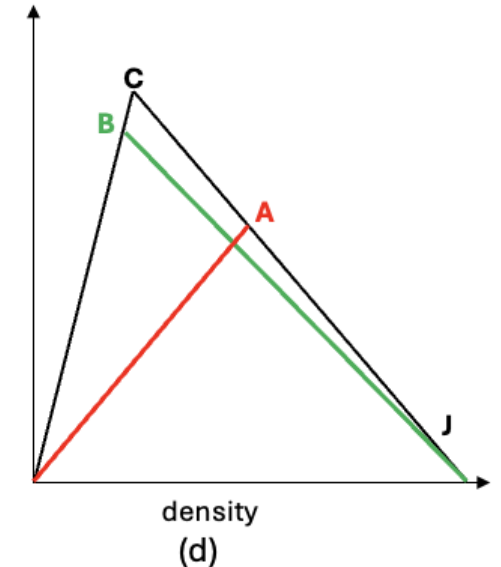
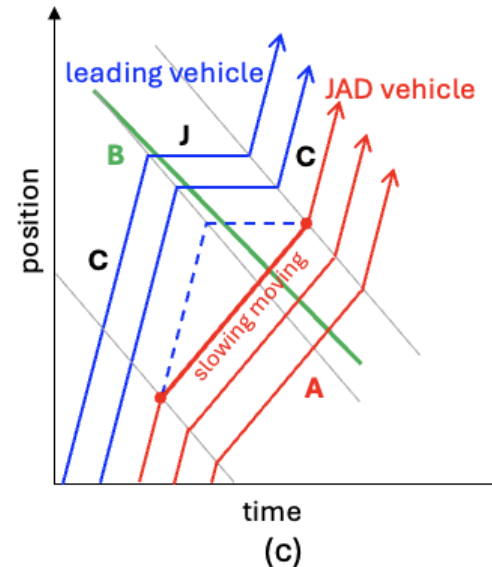
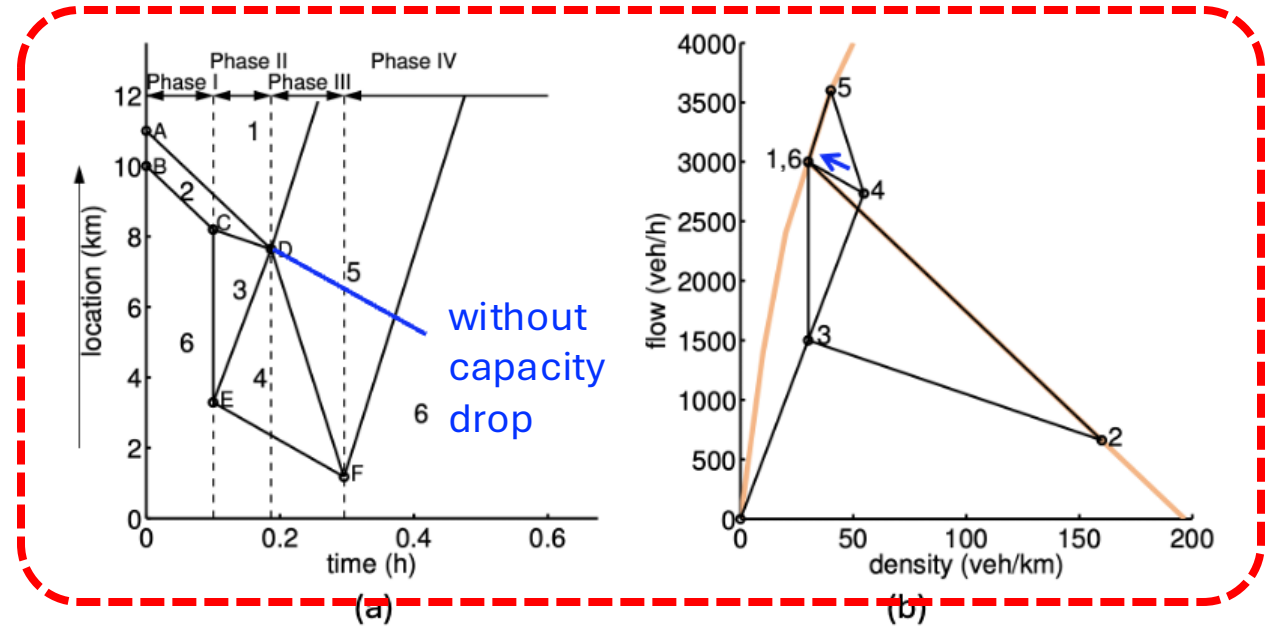
- Most strategies and analyses are based on the FDs or their Lagrangian variants (e.g., speed-spacing relationship). **The shape of the assumed FDs directly influences both the formulation of a strategy and its effectiveness.**



Fundamental Diagrams

VSL

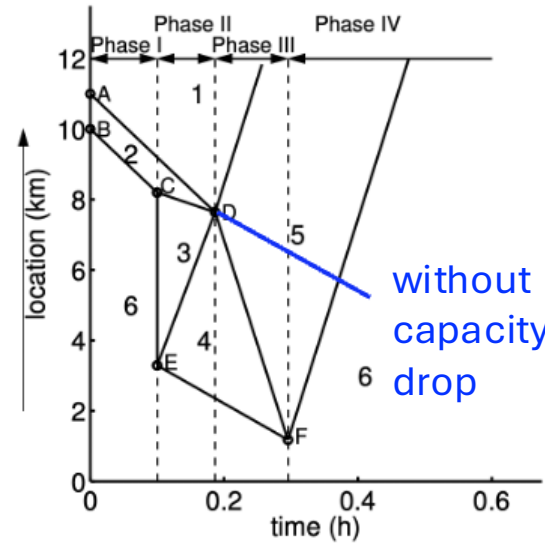
- It is commonly assumed that a **capacity drop** exists
- This assumption effectively resolves the contradiction between the widely observed **constant length** of the moving queue and **its shrinking** after imposing a lower speed limit.
- State 5 is a highly unstable state**, where any disturbance can lead to a traffic breakdown. This indicates that, even though a stop-and-go wave is suppressed, it remains **uncertain how long the high-flow region (state 5) can be sustained**.



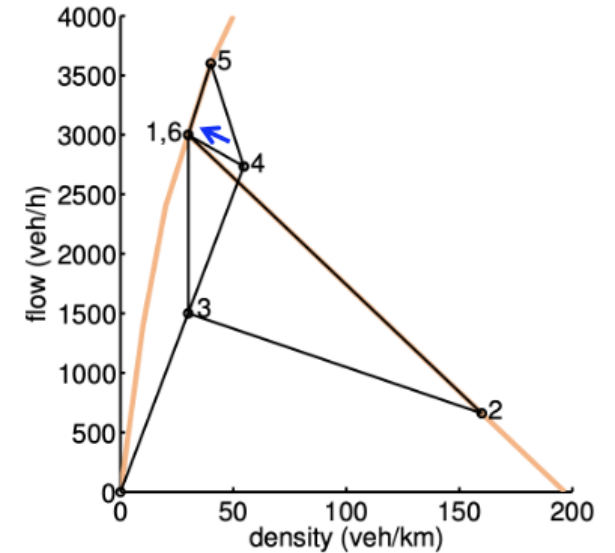
Fundamental Diagrams

JAD

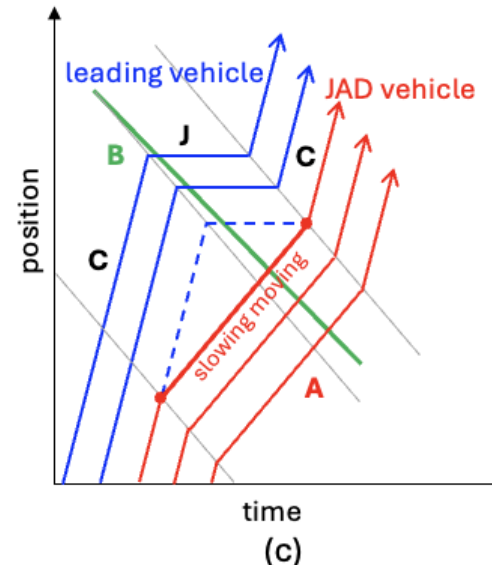
- Most JAD strategies assume TFD
- For example, state A in Figure (c) is still **self-sustained**.
- To achieve a self-shrinking JAD-impacted region, {JAD2013} concluded that an effective JAD requires **the inflow to be lower than the capacity** in TFD (e.g., inflow at state B in Figure (d)). But the wave will shrink by itself



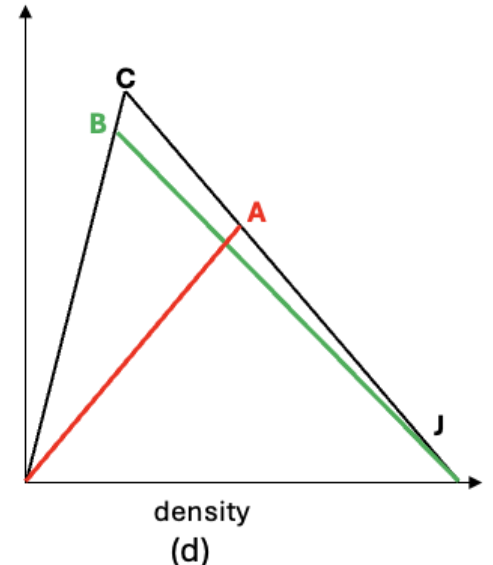
(a)



(b)

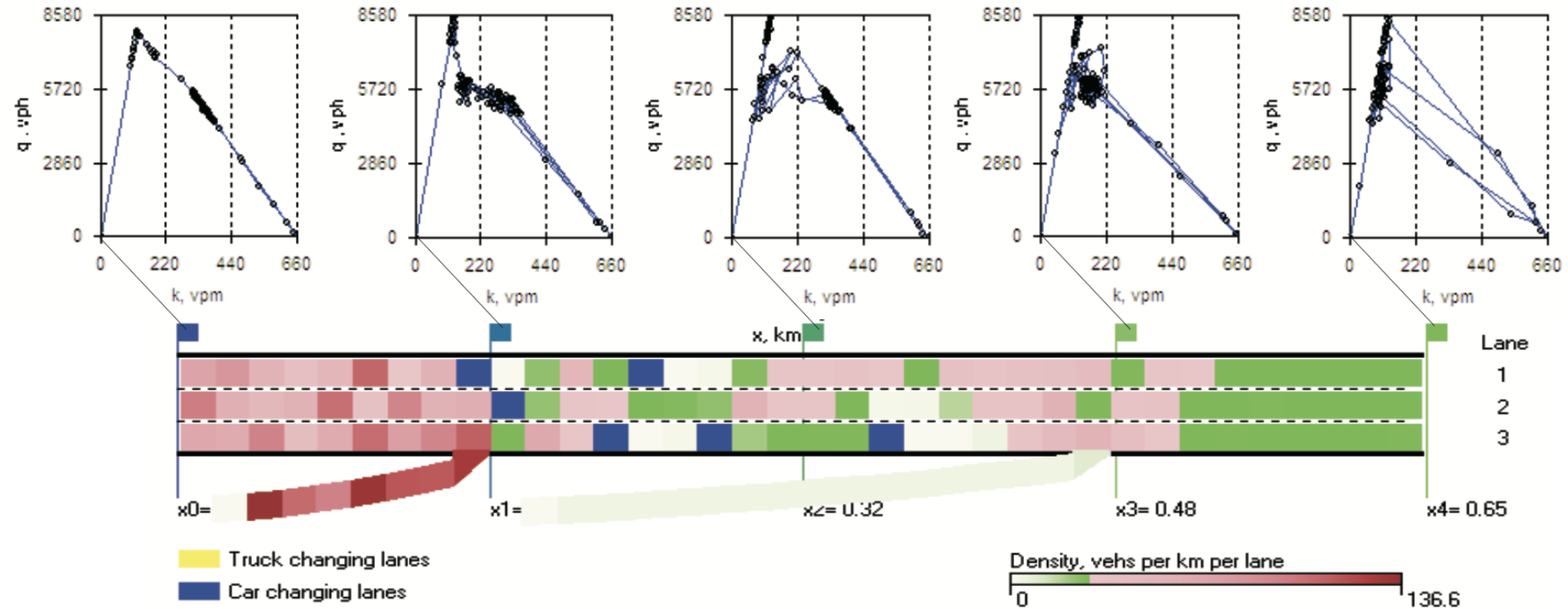


(c)

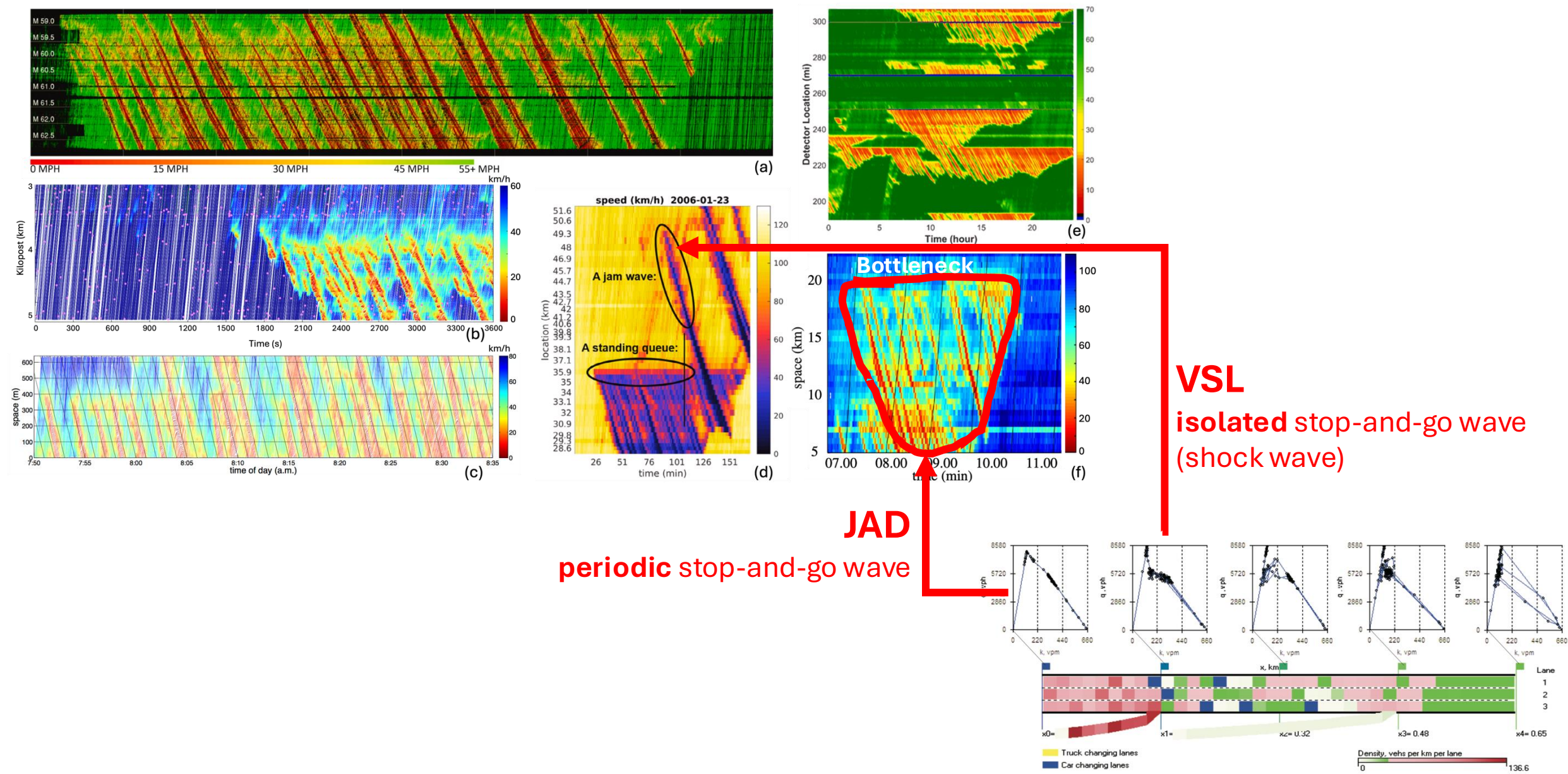


(d)

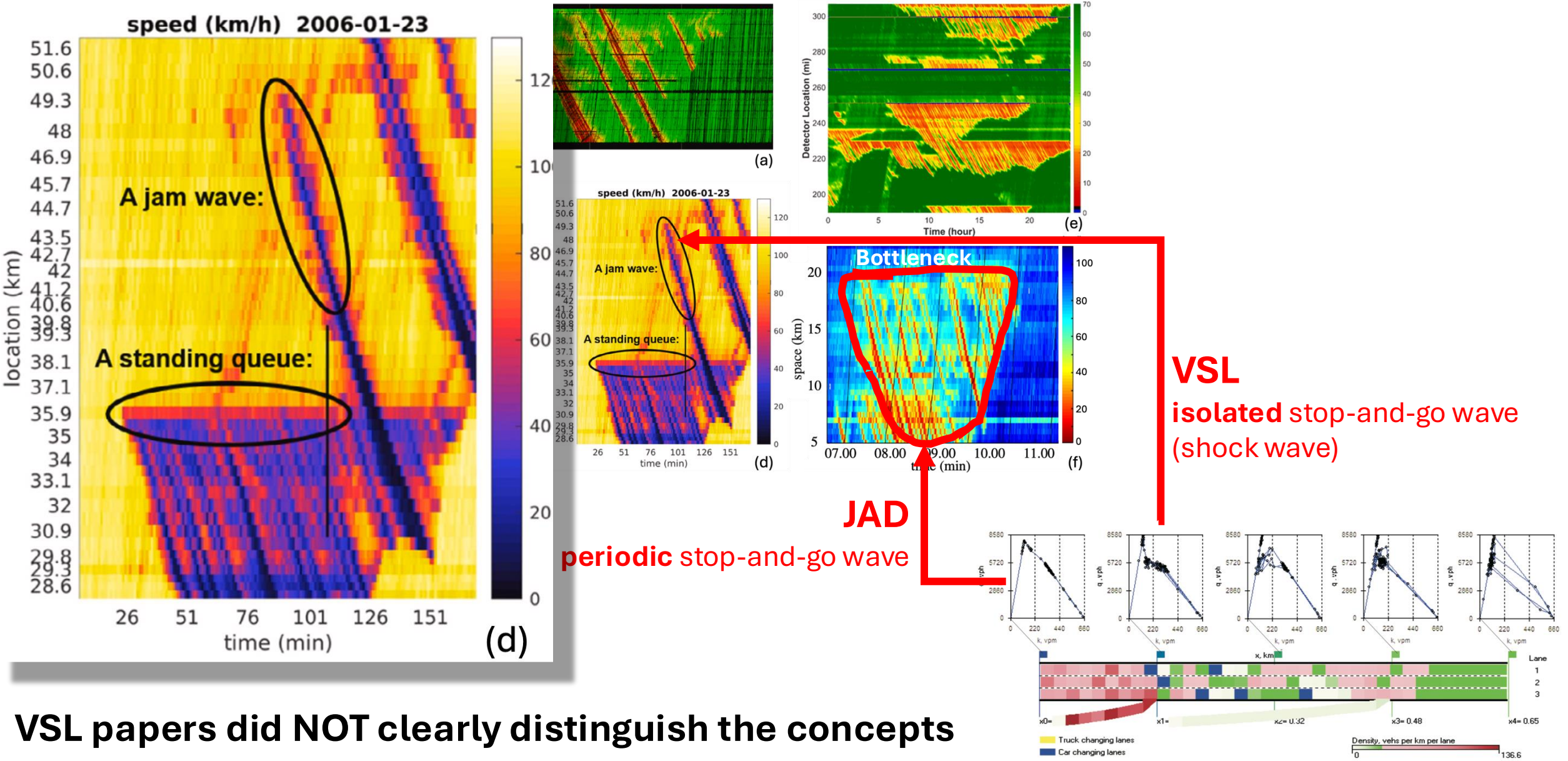
Fundamental Diagrams



Fundamental Diagrams



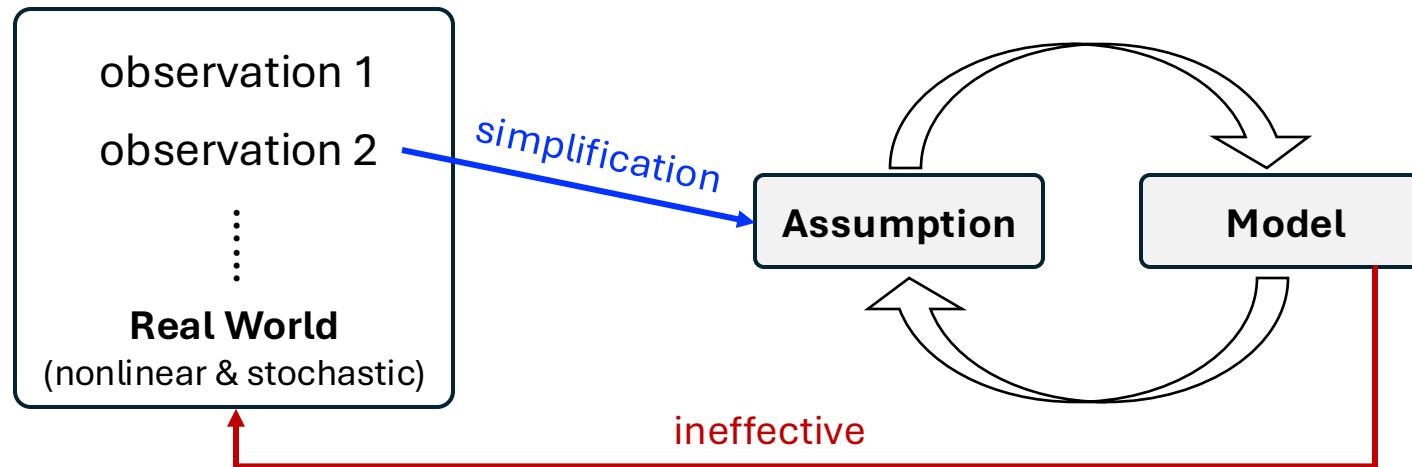
Fundamental Diagrams



VSL papers did NOT clearly distinguish the concepts

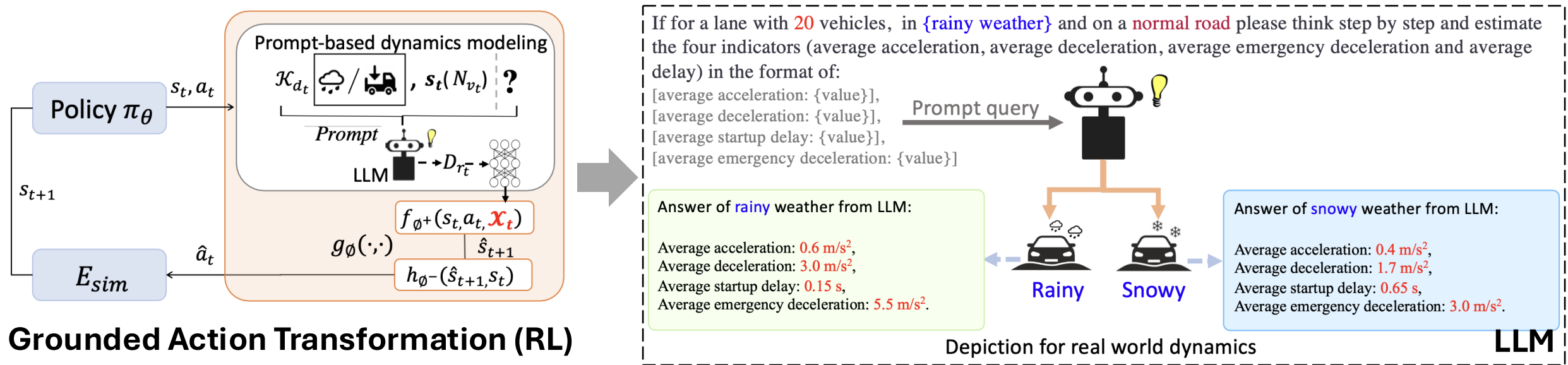
Traffic Dynamics Modeling

- Our researchers are focused on ensuring **internal consistency** in our assumptions and models. Developing these models requires **simplifying certain assumptions and scenarios**, which, in turn, limits their effectiveness in real-world applications due to the inherently nonlinear and stochastic nature of real-world traffic systems.
- Unlike other areas of transportation research, such as those centered on understanding traffic mechanisms or identifying general patterns, **traffic control directly interacts with the system**, placing particularly high demands on the robustness and adaptability of the models.



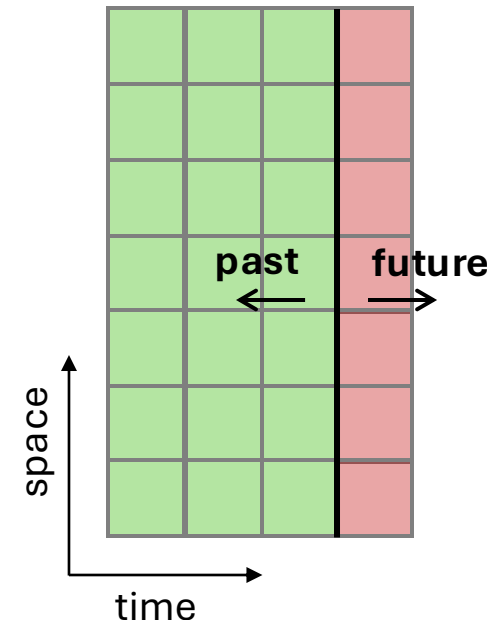
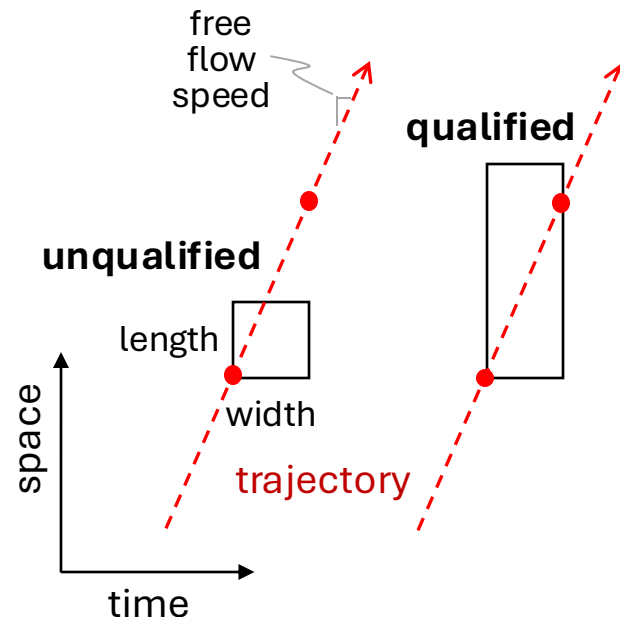
Traffic Dynamics Modeling

- Rapidly-emerged **Generative AI** appears to be a potential tool to mitigate the issues.
- An insightful example might be PromptGAT, which integrates **LLMs to address the Sim-to-Real transfer performance gap** in traffic signal control.
- The model utilizes the reasoning capabilities of **LLMs to generate dynamic information** based on specific conditions, such as weather, traffic status, and road types, which are then applied to enhance the action transformation process in RL models.



Traffic State Estimation and Prediction

- Most of the current **works do not incorporate state-of-the-art** approaches
- For instance, the current MPC-based VSL strategies heavily rely on the prediction of cell-based models such as METANET and CTM. **Their resolution may be too coarse to effectively capture traffic dynamics** in practice (120 km/h freeway: 20 sec * 600 m cell)
- **CNNs** are particularly invented to handle such objects and they have demonstrated the effectiveness in traffic prediction. **Physics-informed machine learning** is also a promising future direction



Stochasticity

- **In proposing a strategy**

Current tools for modeling are all deterministic, while traffic is (highly) stochastic

- FD, KWT → Stochastic Extension
- Car-following Model → Randomize parameters
- MPC → Robust Control, Stochastic MPC

- **In validating a strategy**

- Simulation model → Randomize parameters or data-driven approaches

a) *Probabilistic relation:* In high-density traffic conditions, all vehicles are assumed to be in a car-following state. For this study, we have chosen a **Newell's simplified car-following model**, which is known for its simplicity and universality [22], to describe the behavior of HVs. The model's speed-spacing relation is defined as follows:

$$v = V(s; v_f, l, h) = \min\left\{\frac{1}{h}(s - l)^+, v_f\right\}, \quad (4)$$

where $(.)^+ = \max\{., 0\}$. s represents the spacing and v represents the velocity. v_f, l and h represent the free-flow speed, minimum headway spacing, and expected gap headway, respectively. To account for the heterogeneous behavior of HVs, characterized by the randomness of HV spatial distribution, **we treat the parameter tuple (v_f, h, l) as random variables**. The stochastic speed-spacing relation is then given by

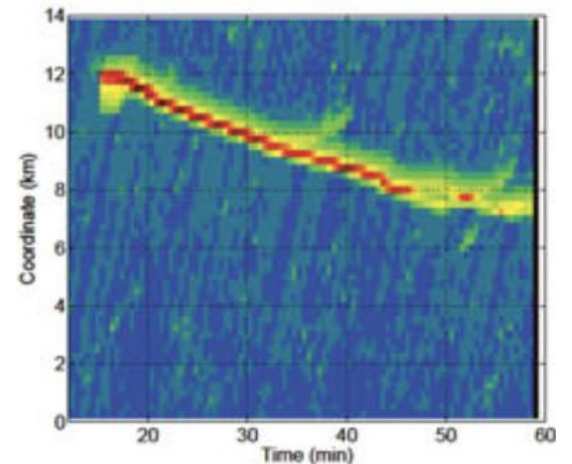
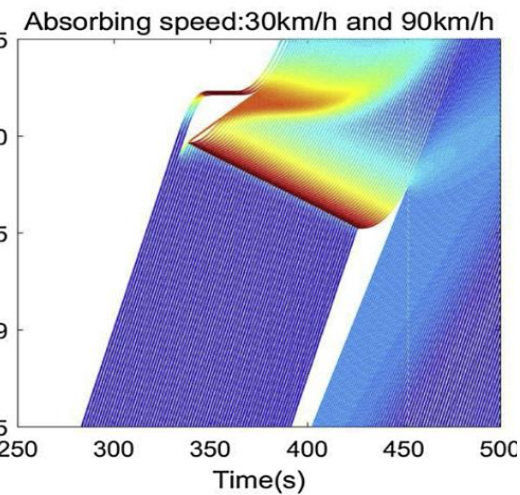
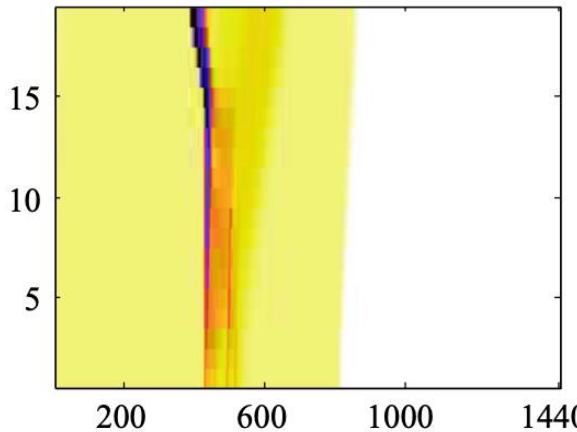
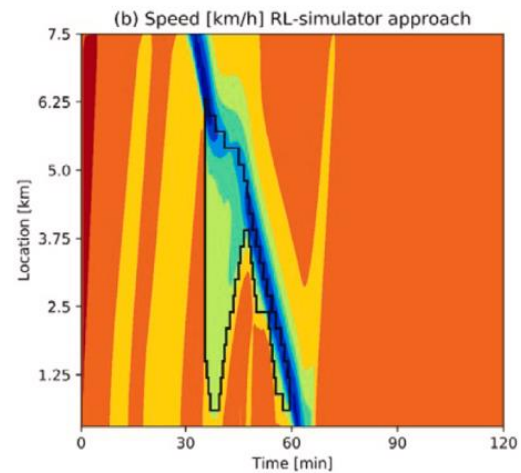
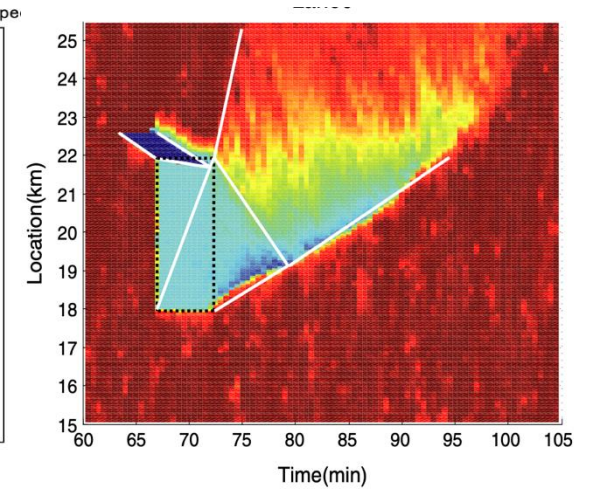
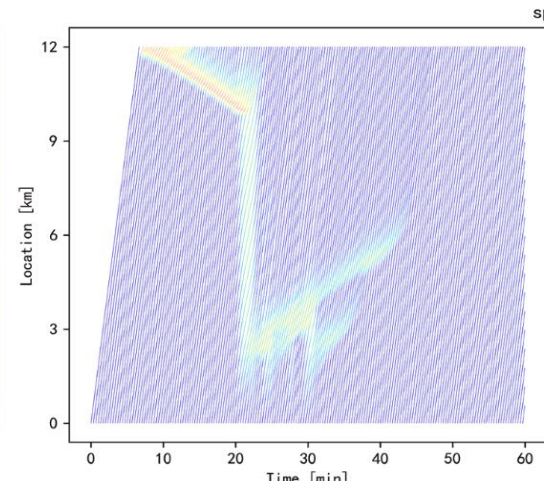
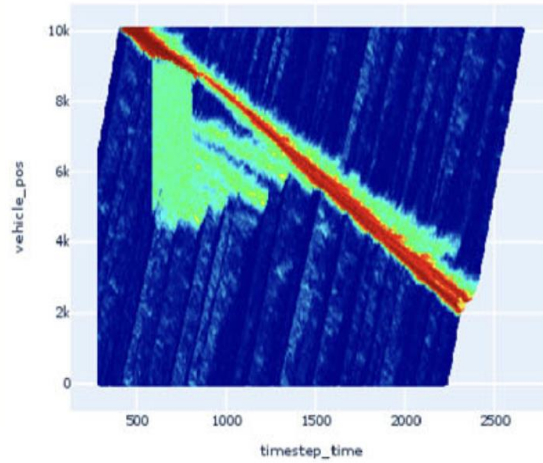
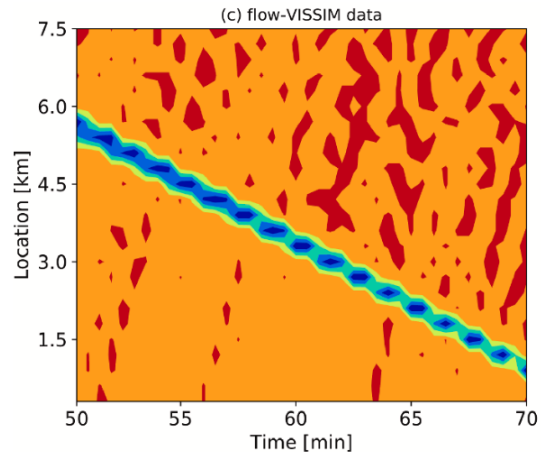
$$\begin{aligned} v(\omega) &= V(s; v_f(\omega), l(\omega), h(\omega)) \\ &= \min\left\{\frac{1}{h(\omega)}(s - l(\omega))^+, v_f(\omega)\right\}, \end{aligned} \quad (5)$$

where $\omega \in \Omega$ represents the random space. For simplicity, ω is omitted in the following formulations. In high-density traffic conditions, all vehicles travel in a car-following state, with a driving speed typically lower than the free-flow speed. Therefore, the spacing of vehicle i (an HV) is given by

$$s_i = h_i v + l_i. \quad (6)$$

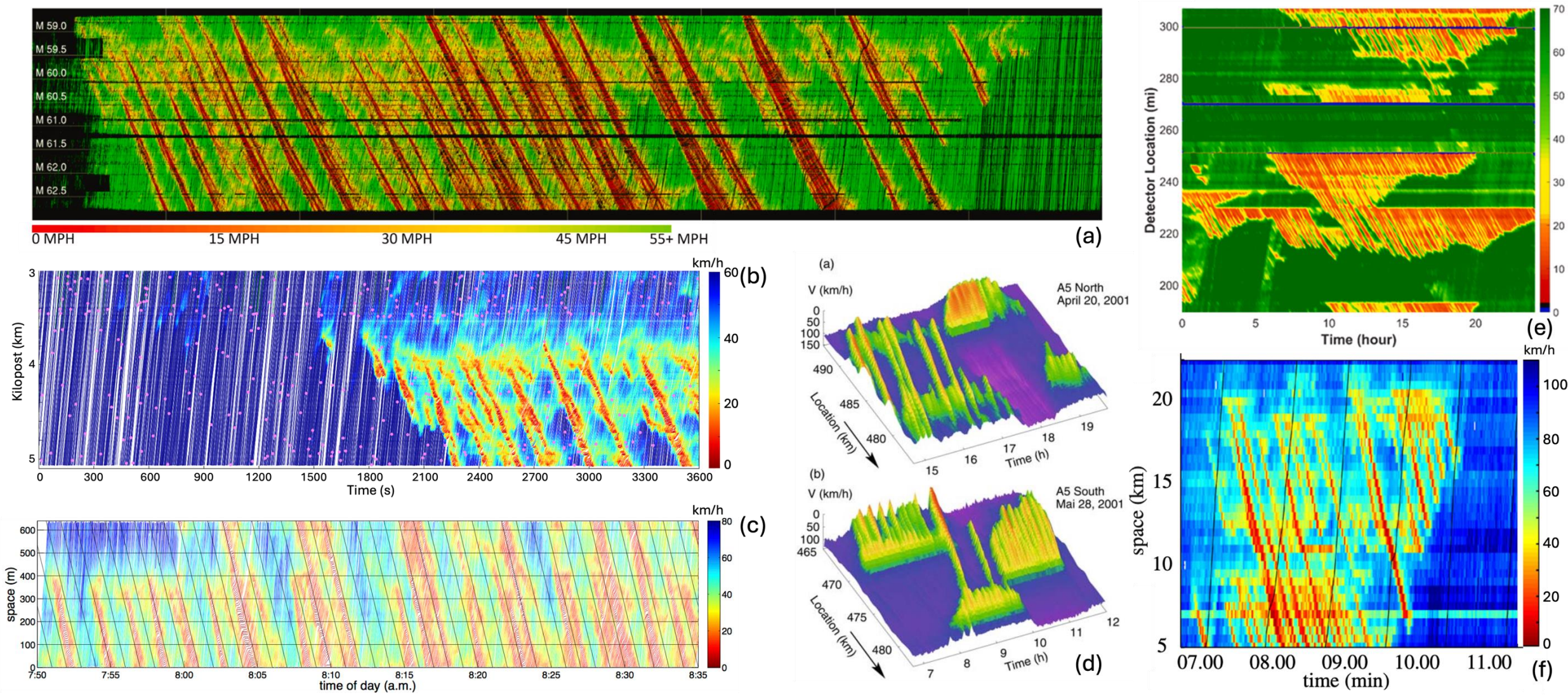
Scenarios for Strategy Validation

- **Insufficient:** Most of strategies (particularly VSL) were tested with a **single stop-and-go wave**



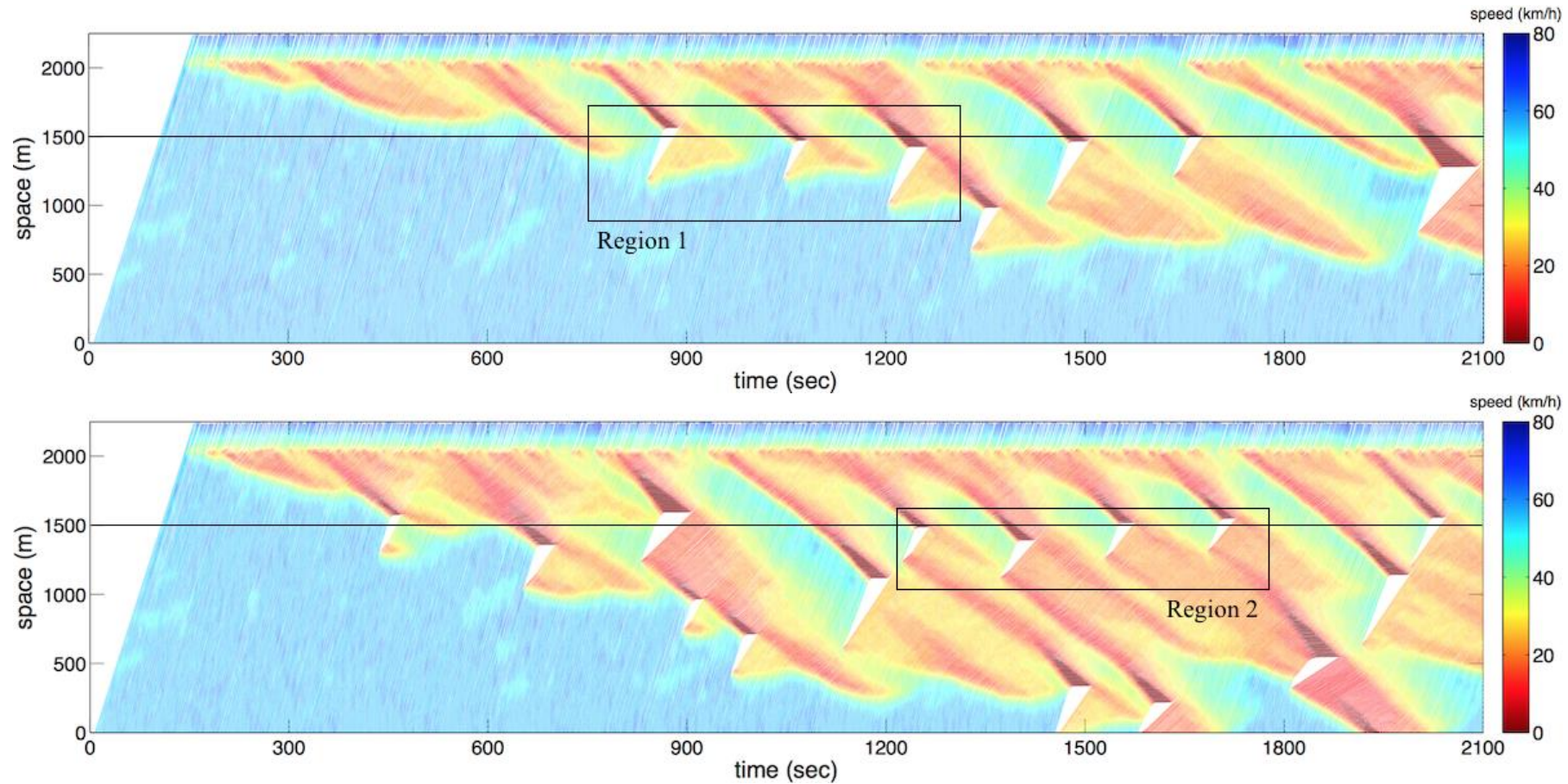
Scenarios for Strategy Validation

- Real world



Scenarios for Strategy Validation

- Multiple wave scenarios

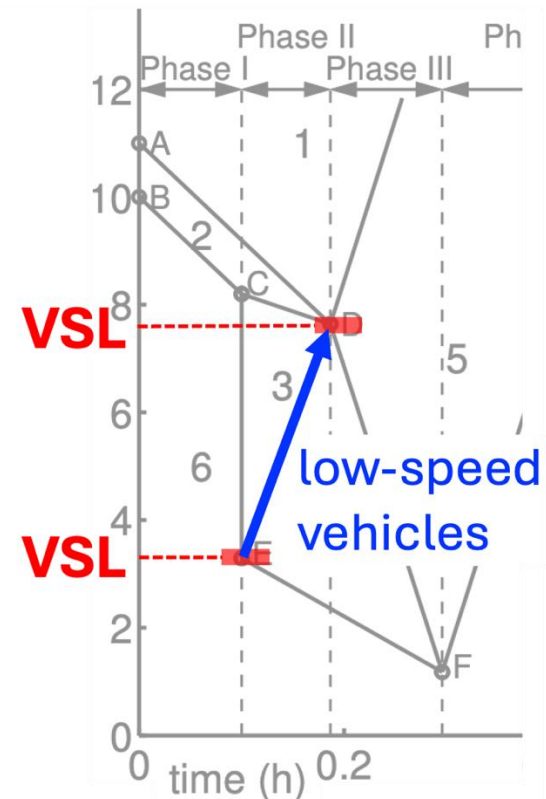
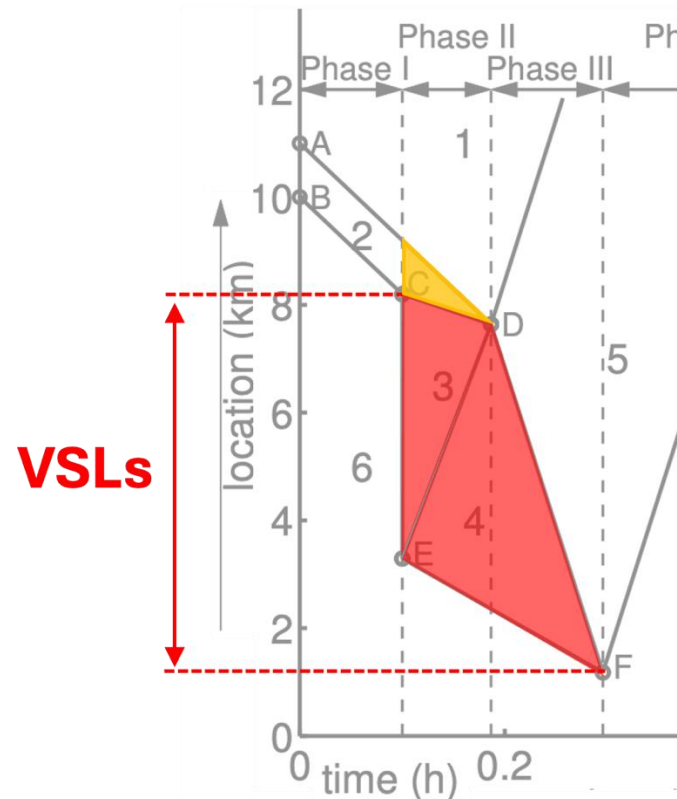


- Trustworthy simulation

Field Test and Practical Deployment

VSL

- To date, {VSL2010} is the **only** field test: over **20** gantries with VMS along a **14 km** freeway stretch
- Even though the KWT-based strategy {VSL2007} is **relatively simple in theory and modeling**, **coordinately disseminating** the information may remain **overly complex for real-world** deployment
- Call for a **bi-VSL** strategy



Field Test and Practical Deployment

JAD: **more flexible with low-cost**

- JAD is Call for a **strategy dedicated to this scenario**

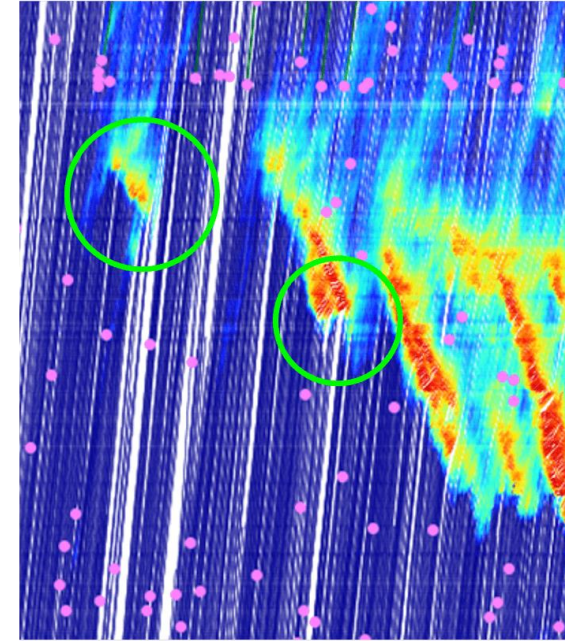
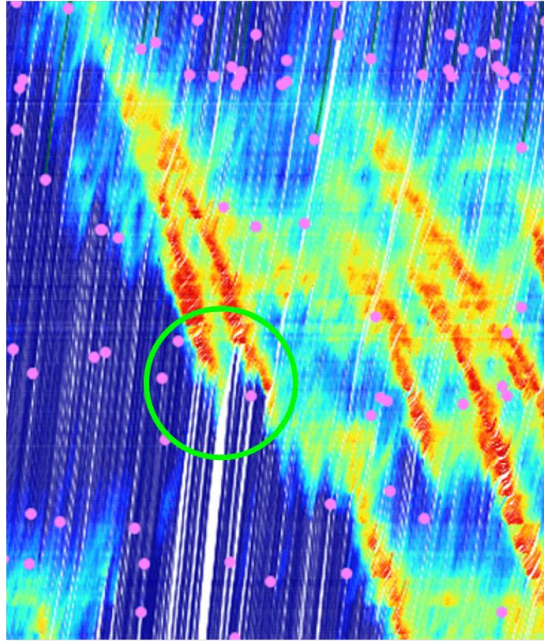
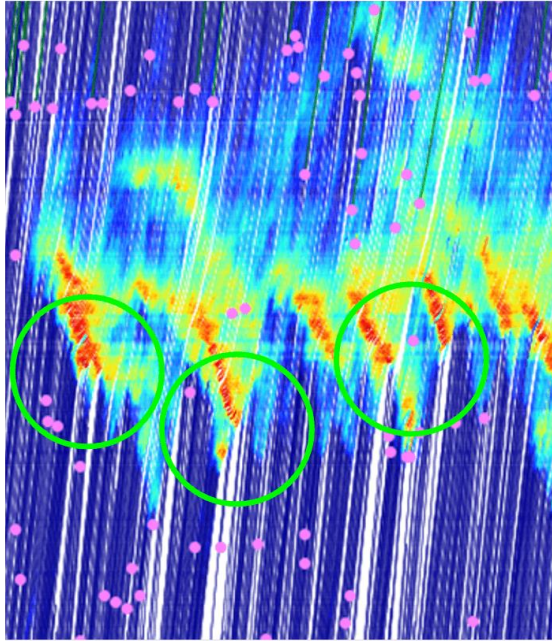


Police car swerving at high speed on a freeway, California, USA

Field Test and Practical Deployment

JAD: more flexible with low-cost

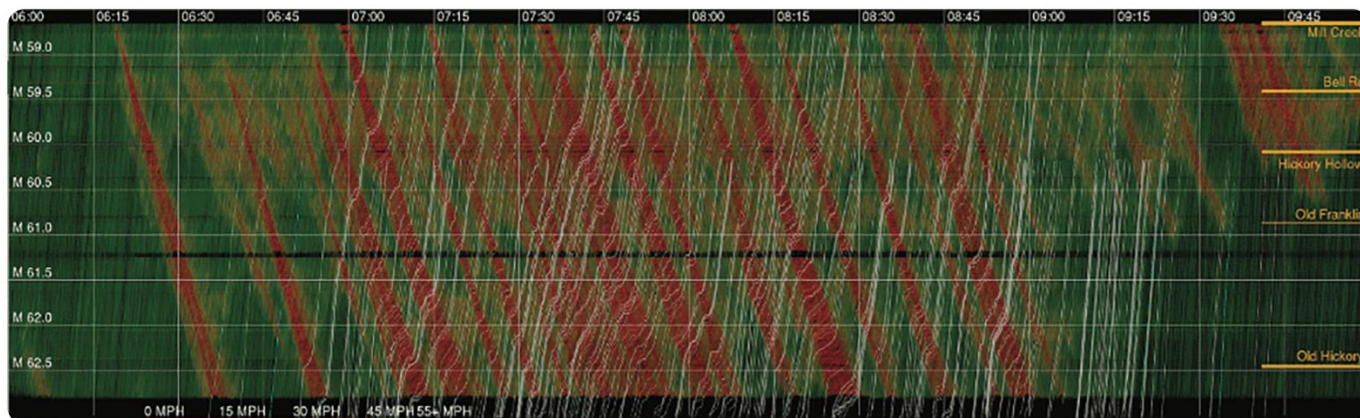
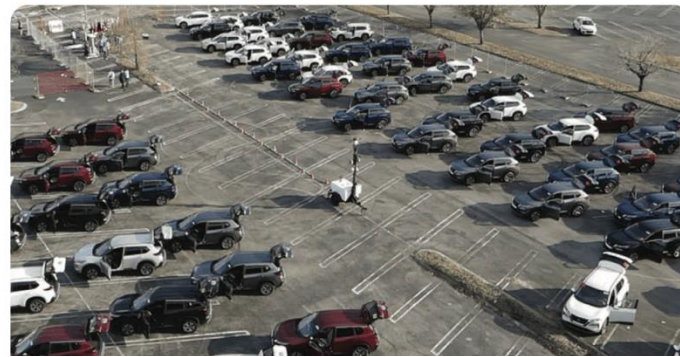
- **Empirical data** suggest that **maintaining an appropriate gap from the leading vehicle** when joining a stop-and-go wave could be effective in suppressing it



Field Test and Practical Deployment

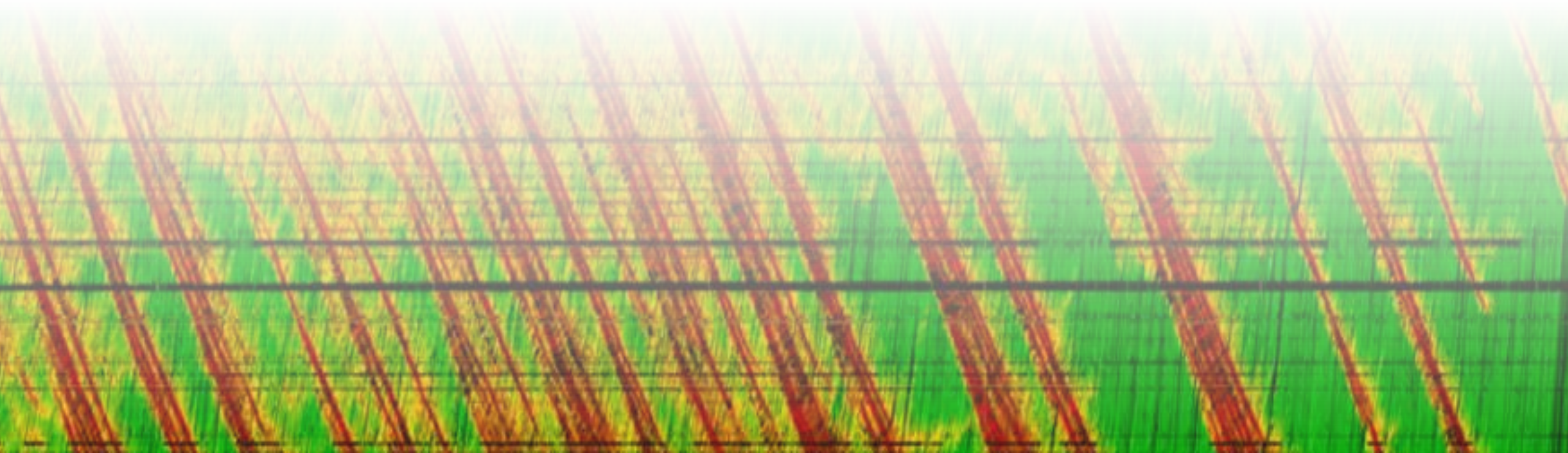
JAD: **more flexible with low-cost**

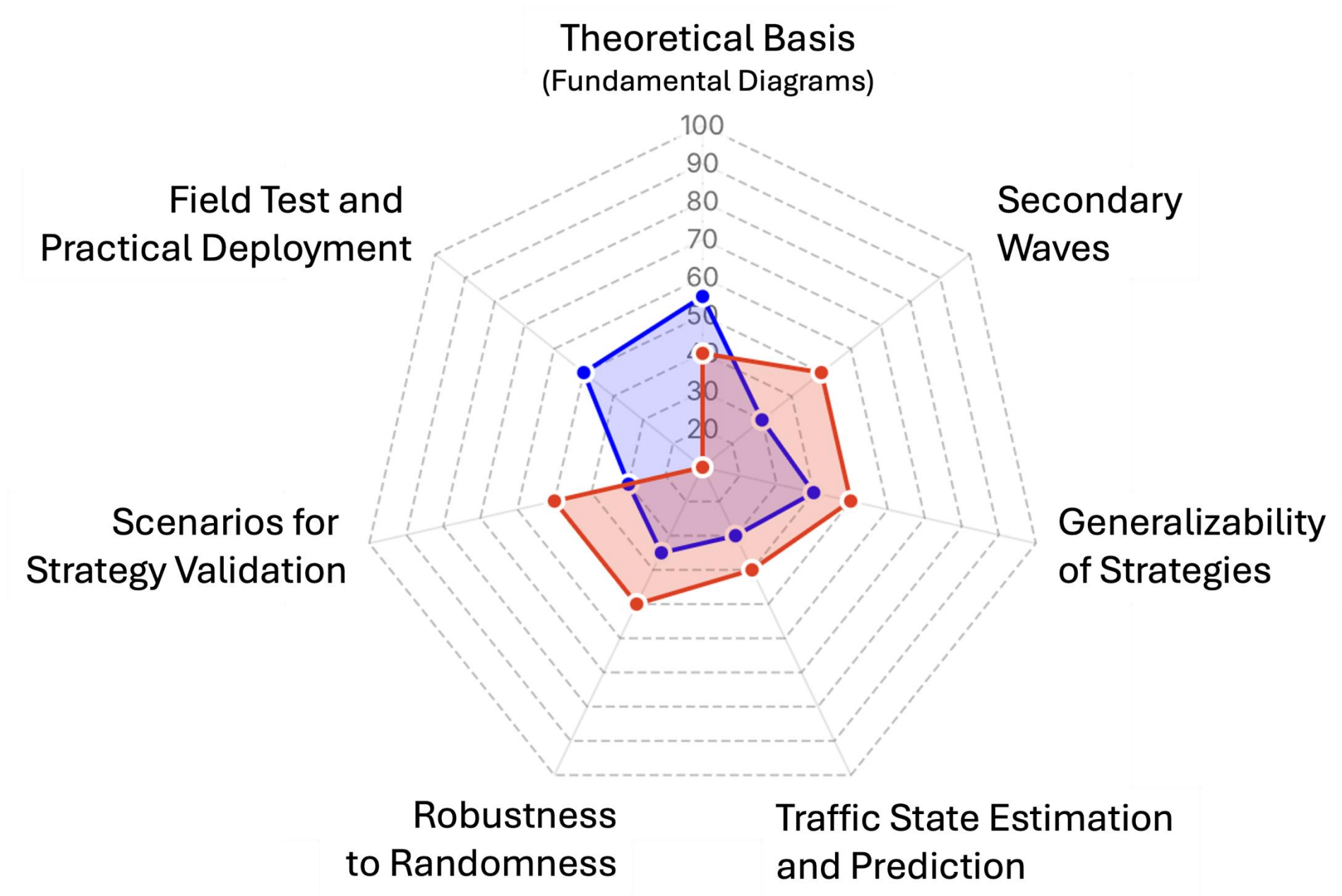
- **The 100 CAV experiment:** highlights the possibility and feasibility of intervening traffic flow through dedicated vehicles even in large numbers, implying the potential of not only JAD strategy but also those VSL-JAV integrated strategies.



Concluding Remarks

Variable Speed Limit vs. Jam-Absorption Driving

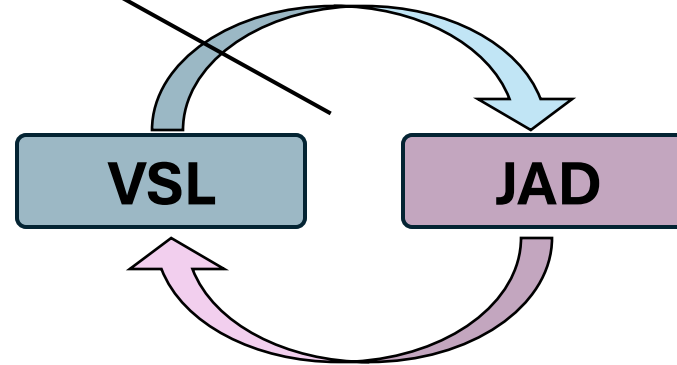




Concluding Remarks

- Lack of solid theoretical foundations: A **lower discharge rate** may be the key
 - **No field test** has been reported: police car swerving on a freeway
 - Making **the inputs more practical** is valuable for a field test

Two areas are merging, particularly in the role of vehicles in intervening traffic, with the rapid development of CAV & V2I



Bi-Level Structure

- Theory: when and why
- Implementation: how

- **Validation** using more realistic (e.g., stochastic) microscopic simulations in multiple-wave scenarios
- The **secondary wave** issue urgently needs to be tested and addressed
- Enhance **traffic estimation and prediction** with more advanced techniques
- Consider more diverse **traffic patterns** and introduce AI tools to reduce the workload
- A **bi-VSL strategy** could be proposed, drawing inspiration from JAD strategies

Thank You

Zhengbing He | www.GoTrafficGo.com

